

Coauthorship integrity: Reconceptualising assessment validity for the age of generative artificial intelligence

Mohsen Ebrahimzadeh^{a,*}, Antonette Shibani^a, Simon Buckingham Shum^b

^a Transdisciplinary School, University of Technology Sydney, Australia

^b Connected Intelligence Centre, University of Technology Sydney, Australia

ARTICLE INFO

Keywords:

Generative artificial intelligence
Coauthorship integrity
Validity
Academic integrity
Non-proctored assessments
Large language models
Dialogic feedback
Interactive orals
Viva voce

ABSTRACT

We consider how a future of pervasive human/AI coauthorship challenges current notions of validity and academic integrity. Specifically, we address widespread concerns that in non-proctored contexts, students are using generative artificial intelligence (GenAI) to submit texts they do not understand. Adopting an assessment validity lens, we show how GenAI undermines the integrity of multiple forms of validity evidence, leading us to propose *Coauthorship Integrity* as a new conceptual source of validity evidence for addressing these threats. Coauthorship Integrity is violated when students submit AI-generated content that they do not understand. To hold students accountable in this regard, a university needs to explore practical methods to gather evidence that could potentially be implemented at scale. We report progress in the development of an “AI Viva”, a conversational agent that engages students in a hybrid experience, combining elements of a viva voce with comprehension questions of controllable type and complexity, engaging the learner with quantitative and dialogic feedback. To provide preliminary expert validation for this prototype, we report qualitative and quantitative evidence from in-depth evaluations with expert educators and assessment experts. By shifting assessment design towards verifying students' epistemic ownership of their writing, this work suggests a potential new pathway for integrating GenAI into pedagogically meaningful, validity-aligned assessment practices. This paper's contributions are thus three-fold: *conceptual* (Coauthorship Integrity), *technical* (LLM-generated questions of controllable difficulty and type, from arbitrary texts) and *empirical* (preliminary expert evaluation of the concept and initial implementation of the prototype's potential for eliciting such evidence).

1. Introduction

Generative Artificial Intelligence (GenAI), powered by Large Language Models (LLMs), has rapidly reshaped how students write, research, and learn. Students can now produce fluent, contextually appropriate text on demand, often with little visible evidence of the cognitive work behind it (Jensen et al., 2025). In non-proctored, take-home assessment contexts, where much university writing remains situated, the ease and invisibility of AI assistance create a pressing challenge: How can assessors be confident that a submission reflects a student's own thinking rather than undetectable outsourcing? This challenge strikes at the heart of assessment validity, which concerns the appropriateness of inferences drawn from assessment scores and relies on multiple sources of evidence (AERA et al., 2014). Historically, written artefacts have served as proxies for the cognitive processes that

produced them. However, the advent of capable LLMs weakens this proxy by decoupling artefact quality from the student's actual cognitive engagement (Perkins et al., 2024), calling into question the validity of inferences drawn solely from product-based evidence.

In this paper we will argue that in higher education, we have now transitioned to a context of ubiquitous human/AI co-creation, whereby AI tools may be used to support any or all of the authorship processes, including problem framing; identifying, summarising and analysing sources; brainstorming and filtering ideas; editing of academic prose (including surface appearance, grammar and cohesion, argumentative and rhetorical structure); and getting critical feedback on drafts. AI-enabled writing support tools are exploding in both design variety and increased availability (Lee et al., 2024). In this new reality, we argue that it is appropriate to move from asking the learner “Is this all your own work?” to “Are you accountable for your work?” Specifically, we propose

* Corresponding author. Transdisciplinary School, Building 7, University of Technology Sydney, Broadway, PO Box 123, Ultimo, NSW, 2007, Australia.

E-mail addresses: Mohsen.Ebrahimzadeh@uts.edu.au (M. Ebrahimzadeh), Antonette.Shibani@uts.edu.au (A. Shibani), Simon.BuckinghamShum@uts.edu.au (S.B. Shum).

<https://doi.org/10.1016/j.caeai.2026.100609>

Received 11 August 2025; Received in revised form 17 April 2026; Accepted 4 May 2026

Available online 23 May 2026

2666-920X/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

the concept of Coauthorship Integrity as a lens through which to explore new strands of assessment validity evidence. In this framing, work is no longer deemed problematic simply because AI was used, but because the student is not accountable for their work, by which we mean they are unable to satisfactorily present, defend, and extend the ideas and reasoning they are developing. This research builds on what we are now learning about university students' varied and nuanced responses to GenAI (Bearman et al., 2025): "A major implication of this work is that educators may need to rethink how academic integrity is characterized when working with GenAI." [...] "A potential direction for further research could involve exploring the relationship between a student's capacity to provide a persuasive rationale for the ideas, sources or claims within submitted work, and the extent to which that work can be said to be owned by that student."

Once we have located Coauthorship Integrity as a plausible response within the assessment validity landscape, the paper turns to the practical implications of making such a shift. While holding oneself accountable for one's work is an academic and professional metacognitive disposition we want to cultivate in students, universities must in turn devise new ways to hold students to account when they use GenAI tools. Thus, as a technical step toward making Coauthorship Integrity a reliable and resource-efficient form of evidence that a typical university could acquire, we report progress in the development of an "AI Viva" implemented as an LLM-based conversational agent that combines elements of a viva voce (i.e., an oral examination for an academic qualification) with text comprehension questions, providing structured feedback, and the option to engage in open-ended dialogue.

Next, we report qualitative and quantitative evidence from in-depth evaluations of this prototype by expert educators, some of whom were also assessment experts. We propose a theoretical basis for Coauthorship Integrity as a source of validity evidence, explore the early technical feasibility of the approach, and provide a preliminary expert validation mechanism for probing student understanding; together, these provide an encouraging proof-of-concept to support a plausible foundation for the further development of an AI viva application for students.

Before turning to the literature, we emphasise that this paper does not present a summative assessment solution, nor does it claim readiness for high-stakes deployment. Instead, the work follows the staged pathway shown in the roadmap (Fig. 1). We begin by identifying how GenAI disrupts established sources of validity evidence in non-proctored writing contexts. Building on this problem analysis, we introduce

Coauthorship Integrity as a conceptual response and position it within the assessment validity framework as a potential new evidentiary source. We then translate this conceptual shift into design requirements for an AI-enabled viva capable of eliciting such evidence at scale. The next step involved developing a prototype that operationalises these requirements through LLM-generated comprehension questions and dialogic interaction. Finally, we report preliminary expert evaluations of the prototype's feasibility and pedagogical value. Together, these stages lay the foundational groundwork for future research on psychometric validation, security considerations, and scalable implementation with students (this paper covers only the steps shaded green).

The paper proceeds as follows: Section 2 introduces key literature to explain how GenAI undermines current forms of conventional assessment validity evidence. In response, Section 3 introduces Coauthorship Integrity as a concept, which may be evidenced through dialogical interaction (with humans or machines). Section 4 specifies the requirements and success criteria for an LLM-enabled "AI Viva" to gather such evidence. Section 5 documents the design process and implementation of the prototype. Section 6 documents the methodology for an expert evaluation study, and Section 7 reports the results. Section 8 broadens the discussion, before Section 9 concludes, recognising limitations, and proposing promising directions for future work.

2. How GenAI undermines established notions of validity

Assessment integrity in the GenAI era must be reframed, moving from a provenance problem (*who wrote the text?*) to a validity problem (*what inferences can be justifiably drawn from the evidence?*). Traditional validity frameworks treat the submitted artefact as the primary evidence of learning, which turns on the premise that the effort required to produce an artefact of this quality is a proxy for a student's effort and understanding. That premise no longer holds. When LLMs can produce high-quality text independently of a student's cognitive work, product-based evidence is weakened. In fact, as explained next, multiple forms of validity evidence are undermined by GenAI.

For about five decades, the authoritative source on assessment validity has been the *Standards for Educational and Psychological Testing*, jointly led by the American Educational Research Association (AERA), the American Psychological Association (APA), and the National Council on Measurement in Education (NCME) (AERA et al., 2014). It follows

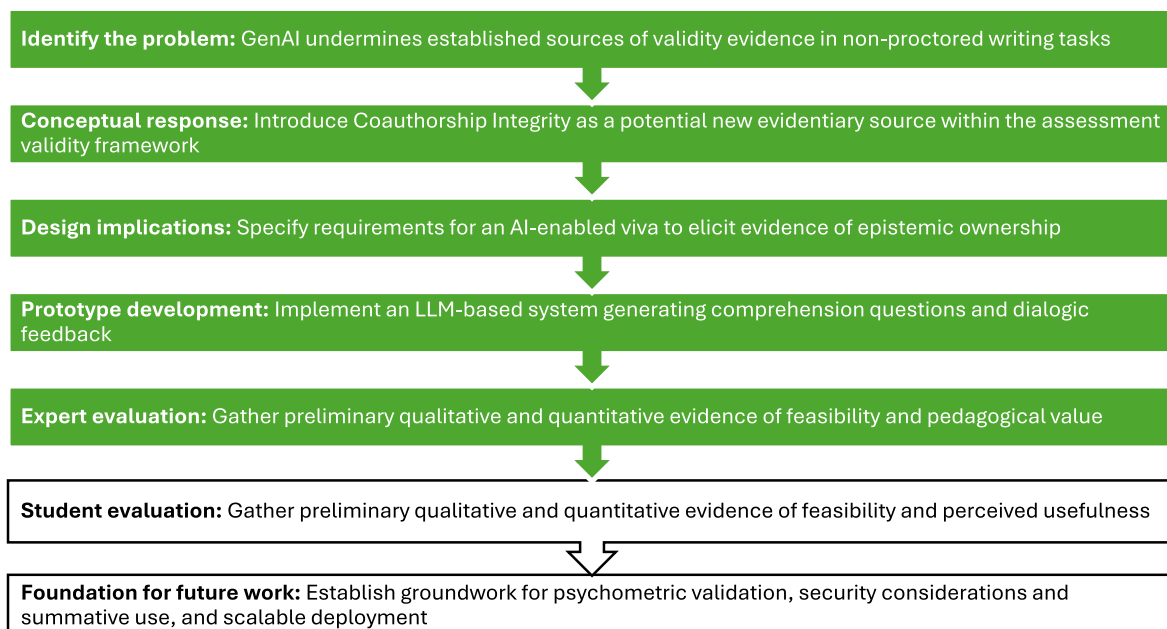


Fig. 1. Roadmap of the paper's staged contribution pathway

Messick's (1989) Unitary Validity Framework, where validity is not a property of tests themselves, but of the interpretations and uses of assessment results. It requires accumulating evidence from multiple sources to support the claim that scores meaningfully reflect the intended construct (e.g., student knowledge, skills, and understanding) rather than extraneous factors. This influential AERA et al. (2014) source identifies five key sources of validity evidence, summarised in Fig. 2 in adapted form, to highlight that student use of GenAI threatens each of these.

AERA/APA/NCME recognises that this framework requires revision in response to the arrival of GenAI. A consultation on the required revisions to the *Standards for Educational and Psychological Testing* confirmed that “respondents emphasized the critical need to address the implications of AI in test development and delivery” (Zahner et al., 2025, p. 14). This paper seeks to address precisely that need. We begin by defining each form of validity evidence, the nature of GenAI's threat, and a typical example of how it manifests in higher education.

2.1. Test content

Definition: This source examines how well the test items represent the construct's domain. It includes analyses of item wording, themes, and formats to determine whether the content adequately reflects the knowledge, skills, or traits defined in the conceptual framework. For instance, in an achievement test, items should align with learning standards and cover the full scope of intended content. Expert judgment often plays a role here; subject matter specialists could assess whether the test captures critical components and avoids irrelevant or biased elements. Content evidence is particularly vital in settings like curriculum alignment, licensure, or certification. Critically, while the prompts and questions may still align well with the construct domain, the issue here is whether the student-generated responses continue to reflect the

intended knowledge or skill.

GenAI's validity threat: If GenAI tools embellish the student's work with advanced vocabulary, complex sentence structures, or external content the student does not understand, it obscures the link between item content and student ability. The assigned task remains valid, but the responses no longer provide trustworthy reflections of student performance on that content. As explained shortly, a key innovation that we demonstrate is the use of the student's own (potentially coauthored) writing as the text content, through the dynamic generation of comprehension questions about this text.

What this looks like in typical HE practice: In a common university assignment such as a short, written explanation of a physiological process in a first-year health subject, a student may submit a well-structured paragraph describing, for example, the renin–angiotensin system. On the surface, the content appears to align perfectly with the intended learning outcomes: the terminology is precise, the sequencing is correct, and the explanation is coherent. However, if the student relied heavily on GenAI to generate or refine this paragraph, the polished response may exceed their actual understanding. When later asked in class to explain the same process verbally or apply it to a clinical scenario, the student may be unable to do so. In this situation, the task still reflects the intended content domain, but the student's response no longer provides trustworthy evidence of their knowledge. The apparent mastery is an artifact of AI-assisted writing rather than the student's cognitive grasp of the material.

2.2. Response process validity

Definition: This source investigates how test takers engage with tasks; for instance, do their strategies and thought processes reflect the intended construct? If a test is designed to measure problem-solving, for example, are the students actually solving problems or just following



Fig. 2. GenAI threat to validity evidence (based on AERA et al., 2014).

memorised procedures? Evidence here includes think-aloud protocols, analysis of draft revisions, eye-tracking, response times, and interviews. It also applies to observers and scorers; to exemplify, do assessors/teachers/judges follow scoring rubrics consistently and focus on the construct rather than irrelevant factors? When the validity argument involves cognitive operations or psychological mechanisms, response process evidence helps ensure the scores reflect those as intended. It is self-evident that in a typical university setting, educators cannot gather fine-grained process data of the sort studied in assessment research labs. However, although a focus on process is not new (Hager & Butler, 1996; Murray, 1972), there is a growing emphasis in the literature on *process-centric* assessment (Gagich, 2025; Smith & Francis, 2024; Winne, 2023), and in addition, the potential for well-designed learning analytics to provide process evidence for critical reflection by educators and students (Shibani et al., 2024), and complex forms of embodied teamwork (Echeverria et al., 2025).

GenAI's validity threat: With the emergence of GenAI, there are justifiable concerns that students have neither learned (nor in most cases received any guidance on) how to orchestrate their own effort with AI to support writing-related tasks spanning research, ideation, drafting, organisation, or refinement. Accordingly, in non-proctored contexts, any behavioural evidence (e.g., time taken, revision steps, or drafting strategies) no longer represents unaided reasoning/writing, but the human/AI “pragmatic system” (Pea, 1985). Consequently, interpretations based on response quality become less defensible.

What this looks like in typical HE practice: As an example, a lecturer might assign a short reflective task asking students to explain how they approached a complex reading or how they developed an argument in an essay. Before GenAI, the lecturer could reasonably assume that the student's description of their process (e.g., how they selected sources, why they structured their argument in a particular way, or what steps they took to revise their draft) reflected their own cognitive engagement. With GenAI, however, a student may simply ask an AI tool to “explain how this essay was written” or “summarise the reasoning behind this argument” and then submit that explanation as if it were their own. The lecturer receives what appears to be a thoughtful account of the student's process, but it no longer provides genuine insight into the student's reasoning or decision-making. The behavioural evidence that once helped educators infer how a student engaged with the task is now entangled with AI assistance, making it difficult to determine whether the student actually performed the intended cognitive work.

2.3. Internal structure validity

Definition: This source evaluates the relationship between test items and components to see if they conform to expectations based on the construct of interest. Techniques like factor analysis, item correlation matrices, and dimensionality assessments are used to determine whether items form coherent scales or distinct subscales (subdomains). If a test claims to measure multiple factors (e.g., verbal and quantitative reasoning), the data should reflect that structure. Internal structure evidence also includes analyses of differential item functioning; for example, examining whether items behave differently for comparable subgroups (e.g., across gender or linguistic backgrounds) may suggest validity or fairness concerns.

GenAI's validity threat: Diverse student use of GenAI in unverifiable ways introduces construct-irrelevant variance, making item correlations less meaningful. If multiple tasks are used (e.g., reflective writing, persuasive essay), their internal relationships may blur because AI's stylistic and structural influence can homogenize responses across tasks and students. This undermines the ability to assess latent dimensions like reasoning, coherence, or originality through factor or dimensionality analysis.

What this looks like in typical HE practice: In a typical subject, students may complete several written tasks across the semester such as

a reflective journal, a case analysis, and a short essay each designed to tap into different subcomponents of the intended learning outcomes. Before GenAI, educators could usually detect meaningful variation across these tasks: a student who excelled in structured argumentation might struggle with reflective writing, while another who wrote fluently in narrative form might find analytical tasks more challenging. These differences helped instructors infer the underlying dimensions of student ability and supported coherent judgements about progress. With GenAI, however, students may use the same tool to generate or refine all three submissions, producing work that is uniformly polished, stylistically consistent, and rhetorically similar across tasks. The natural variation that once revealed distinct subskills is flattened, making it difficult to determine whether the student genuinely possesses the differentiated abilities the tasks were designed to assess.

2.4. Relations to other variables (criterion-related validity)

Definition: This evidence supports propositions that interpret test scores as indicators of future behaviour or traits. Do assessment results correlate with similar constructs (convergent evidence) or avoid correlating with unrelated ones (discriminant evidence)? Do they predict relevant outcomes (criterion-related evidence) such as academic success, job performance, or clinical indicators? Do the results transfer to relevant but different settings? (This can also be examined through literature meta-analysis.) Clearly, any such validity claims are compromised if the scores do not stem from the students' own reasoning or effort.

GenAI's validity threat: AI-generated work may mimic high-achieving writing which could artificially inflate scores and distort correlation patterns (e.g., with GPA, peer review outcomes, or instructor judgments). This weakens both convergent and predictive validity, as the scores no longer represent the construct consistently across populations or over time.

What this looks like in typical HE practice: A common example arises in subjects where written performance is assumed to predict later success such as a nursing student who submits an excellent written care plan early in the semester. The care plan may be well-structured, clinically appropriate, and written in a professional tone, leading the educator to infer that the student has strong clinical reasoning skills and is likely to perform well in subsequent simulations or placements. However, if the student relied heavily on GenAI to generate or refine the care plan, the high score no longer reflects their actual competence. When the same student later participates in a simulation requiring real-time decision-making, they may struggle to articulate rationales, prioritise actions, or adapt to changing conditions. The expected correlation between the written task and later performance breaks down not because the assessment was poorly designed, but because the written evidence no longer originates from the student's own reasoning.

2.5. Consequences (consequential validity)

Definition: The final source considers the validity of using test scores to make consequential decisions, which may range from assigning a grade, to continued enrolment, admission to a new program, or graduation. Equally important are *unintended* consequences, such as curriculum narrowing or adverse impacts on specific groups. If negative outcomes result from construct underrepresentation or construct-irrelevant variance, they may indicate a validity problem. Moreover, consequences that lie outside technical validity (e.g., public perception or motivational effects) may influence decisions about test use or reduce their trustworthiness within society more broadly. An example would be concern about academic pressure on high stakes exam performance plausibly associated with student mental health problems (Stear et al., 2023). Consequential validity thus prompts reflection not only on the kind of decision that a score legitimises, but the broader effects of test design and implementation.

GenAI's validity threat: The arrival of GenAI calls into question the evidential basis of consequential decisions, most profoundly, whether a student deserves certification. GenAI may also precipitate unintended consequences, for instance, students might become over-reliant on external tools, atrophy of metacognition (Fan et al., 2025), bypass deep learning or self-reflection, and ultimately diminish the educational impact of writing tasks. If instructors react by tightening controls or abandoning take-home tasks altogether for proctored exams, this shrinks instructional variety and assessment authenticity. In turn, such measures may unfairly disadvantage certain student groups.

What this looks like in typical HE practice: A common occurrence of this is when written assessments are used to make meaningful decisions about student progression. Consider a capstone subject in which students submit a major written report that contributes heavily to their final grade and determines eligibility for placement or graduation. A student may submit an exceptionally polished report that is clear, well-structured, and professionally written leading the educator to infer that the student is ready for more advanced responsibilities. However, if the student relied extensively on GenAI to generate or refine the report, the high grade may not reflect their actual capability. When the same student later enters a placement, they may struggle to apply the concepts they demonstrated in writing previously or be unable to justify decisions that appeared well-reasoned in the report. The consequential decision (e.g., passing the subject, progressing to placement, or graduating) is therefore based on evidence that does not originate from the student's own understanding. At scale, this can erode trust in assessment outcomes, widen inequities, and prompt institutions to adopt more restrictive assessment practices that may disadvantage certain groups of students.

2.6. Prior work on LLM-generated comprehension questions

Given our focus on Coauthorship Integrity, the key point of distinction is that the AI Viva uses the student's own submitted text as the primary source for item generation. This contrasts with a range of studies validating the technical efficiency and psychometric soundness of LLM-generated tests grounded in curricular materials. Isley et al. (2025) conducted one of the largest field studies to date across 91 college courses, demonstrating through Item Response Theory that AI-generated questions performed comparably to expert-created items in terms of difficulty and discrimination. Similarly, Elkins et al. (2024) found that teachers often preferred AI-assisted quizzes, suffering no loss in quality compared to handwritten versions. Lodovico Molina et al. (2024) demonstrated that advanced LLMs can extract granular concepts from university slides to generate relevant questions without fine-tuning. Badenes-Olmedo and Eyzaguirre (2026) introduced QuerIA, which generates and evaluates Bloom-aligned questions from academic materials from instructional documents, while Salminen et al. (2024) developed Cipherbot, a Retrieval-Augmented Generation system that answers student queries strictly from course textbooks. Duong-Trung et al. (2024) proposed BloomLLM, fine-tuning models to ensure semantic interdependence across Bloom's levels, and Nikolovski et al. (2025) introduced an Alignment Evaluation framework to ensure exam questions reflect course objectives. Wang et al. (2025) introduced structured prompt patterns grounded in a collective knowledge base of teachers' questioning strategies. In practical application, Kituku et al. (2025) integrated AI-generated questions into live lectures and discussion forums to monitor attendance and conceptual understanding in real time.

These empirical results add confidence to our choice of LLM-generated items as an approach, but our focus switches to probing the student's epistemic ownership of their own work. This exemplifies the transition called for by Bearman et al. (2025), towards notions of academic integrity grounded in whether students can meaningfully account for the reasoning and claims embodied in their work.

3. Coauthorship integrity via AI Vivas

Before detailing the design implications of Coauthorship Integrity, we first clarify how this proposal aligns with the Standards' treatment of validity evidence. Coauthorship Integrity as a new source of validity evidence is consistent with the Standards for Educational and Psychological Testing (AERA et al., 2014), which emphasise that validity is a unitary construct supported by multiple sources of evidence rather than a fixed taxonomy of validity types. The five commonly cited sources (i.e., test content, response processes, internal structure, relations to other variables, and consequences) are explicitly described as illustrative rather than exhaustive; they are also not mutually exclusive. The Standards note that validation requires articulating the propositions that must hold for a score interpretation to be justified, and that new forms of evidence may be required when new threats to those propositions emerge (see also Zahner et al., 2025). In this sense, the framework is intentionally open to evolution as assessment contexts and technologies change.

Coauthorship Integrity addresses a proposition that the existing sources do not explicitly consider: Is the student the epistemic agent behind the ideas and reasoning expressed in the submitted work? This question is logically prior to the forms of evidence described in the Standards. Response-process evidence, for example, assumes that the test taker is the author of the response and focuses on the cognitive strategies used while responding to items. In GenAI-mediated writing, however, the primary threat is that the submitted text may not reflect the student's own cognitive processes at all. Furthermore, when authorship is uncertain, the evidentiary basis for all other sources (i.e., content, response processes, internal structure, relations to other variables, and consequences) is compromised as discussed in Section 2. Coauthorship Integrity, therefore, is structurally different and functions as a foundational evidentiary strand that supports the validity argument in contexts where GenAI may obscure the student's epistemic ownership of their work. It complements, rather than replaces, the existing sources by addressing a new threat that the Standards could not have anticipated. Throughout the remainder of the paper, we therefore explore Coauthorship Integrity as a potential new source of evidence that could support the validity argument in GenAI-mediated assessment contexts.

Revisiting the unitary concept of validity introduced in Fig. 2, in an age of pervasive human-AI coauthorship, we propose that this could be extended with a new source of validity evidence, *Coauthorship Integrity*, as shown in Fig. 3. This source provides evidence of the student's comprehension of what we must assume, in a non-proctored context, may be a text coauthored with AI.

Two strands for collecting evidence for *Coauthorship Integrity* are shown, which already exist to provide evidence of a student's ability beyond a written text, and could in principle include any social, professional, empirical, or theoretical practice which gives them the opportunity to evidence their comprehension of the ideas they have written about. *Dialogic Interaction* is currently evidenced in some programs through in-person/online viva voce examinations/interactive orals (we are using them interchangeably), which bring the benefits of human interaction, but are correspondingly limited by the need for trained personnel able to conduct and assess equitable dialogues, coordination logistics, and time-consuming to run especially with larger cohorts of potentially hundreds of students (Davis & Engward, 2018; Kumar et al., 2021). We recognise peer-peer dialogue as another approach, whereby students also take on the role of 'interviewer' and 'educator' giving feedback. A body of research clarifies the advantages and limitations of this approach (Schillings et al., 2021; Zhu & Carless, 2018).

As detailed in the next section, we propose "AI Vivas" as a significant new, complementary assessment instrument that is personalised to each student's writing, yet in principle consistent and scalable within a typical university's resource capability once they can deploy/license secure, stable, LLM-powered tools. We emphasize that Fig. 3 is not an

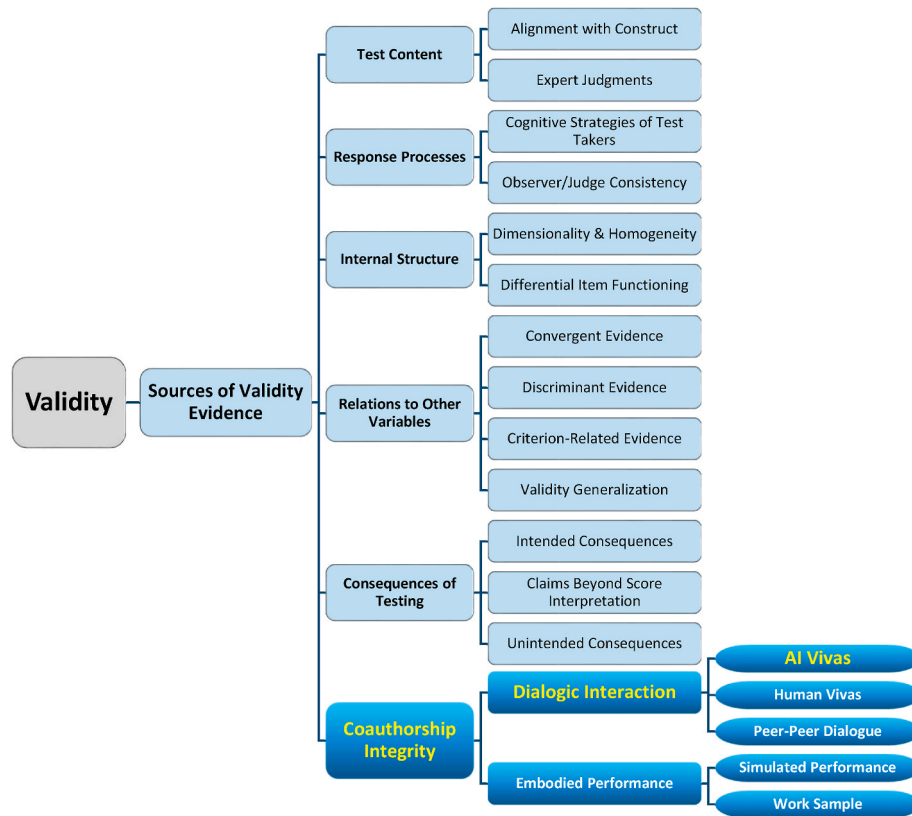


Fig. 3. Conceptualising Coauthorship Integrity as a potential new source of validity evidence, with two exemplar evidence strands, Dialogic Interaction being the focus of this paper. Ellipses show candidate assessment methods, with AI Vivas representing a new method to implement Dialogic Interaction.

exhaustive taxonomy, and for comparison note another strand of evidence that could be used to verify Coauthorship Integrity when the writing concerns embodied competencies (for instance, it is one thing to write about managing a classroom, or the principles of business negotiation, and quite another to do this well). *Embodied Performance* can use educational performance testing (Osterlind, 1998; Waugh & Gronlund, 2013) observed in a real workplace (*Work Sample*), or in a controlled environment like a classroom simulation (*Simulated Performance*), which can now be augmented with multimodal teamwork analytics (Echeverria et al., 2025). Other examples include the Objective Structured Clinical Exam (OSCE) in health disciplines (Zayyan, 2011), or using a pedagogically-designed roleplay chatbot to practice specific communication skills (Lee et al., 2025). Each assessment strategy varies in how specific, structured, hierarchical, or concrete the evidence would be, and thus, their use could be considered in combination with one another depending on the context, as additional convergent validity evidence.

It is important to distinguish between our theoretical proposal of Coauthorship Integrity and the empirical demonstration of the AI Viva mechanism reported in this study. While we argue that the AERA et al. (2014) framework could be extended to include Coauthorship Integrity as a conceptual bridge for addressing GenAI threats, the current study focuses also on evaluating the technical and pedagogical feasibility of the AI Viva as a mechanism for probing student understanding. By reporting the potential of an LLM agent for generating items, scoring, and feedback that experts find educationally useful (Sec. 6), we provide an encouraging proof-of-concept for the AI Viva as a method to gather evidence of Coauthorship Integrity, in response to the GenAI threat to assessment validity.

4. “AI Viva”: Software requirements and research questions

To operationalise the concept of an AI viva, we conceived an interactive software agent that generates a structured dialogue, questioning the student about their work in a variety of ways. While advances in voice recognition and synthesis make AI simulation of an oral viva technically possible, our primary goal is not to emulate the human viva in every respect. The social dynamics of the interview bring the social pressure to respond rapidly, demonstrating agility of thinking and verbal dexterity. While this ability is critical to cultivate in many educational contexts, it is not our primary interest in establishing Coauthorship Integrity (and moreover, this format can disadvantage students not fluent in the language, or who have other accessibility needs). The verbal format also rules out the ability to pose challenging questions of certain sorts that would be too complex to introduce in a conversation (due to working memory load), and are much better presented on screen (as detailed in Sec. 5.3). Our AI viva is thus a hybrid experience, simulating *specific elements of a well conducted human viva* (a 1-1 dialogue; carefully prepared questions of known difficulty designed to test the student's ability to explain their work; consistent feedback and grading; an open-ended invitation to discuss anything at the end), but the presentation of those questions uses the *visual affordances of on-screen comprehension items*, enabling *students to respond via the visual user interface* (keyboard and item selection).

The functional requirements for the first prototype agent were therefore specified as follows:

- Accept an arbitrary text from the user
- Generate robust comprehension questions of different types from this text, optionally specifiable by the teaching team.

- Where relevant, vary these comprehension items to probe more than one level of Bloom's revised taxonomy (Anderson et al., 2001) of varying difficulties
- Assess the consistency of the user's responses with their text
- After all questions have been answered, generate feedback on performance
- Invite the user to engage in dialogue about the results or any other aspect of the viva

Any use of AI to generate test items must satisfy quality criteria that are agnostic to the domain (Martin Kowal et al., 2025; Rodriguez & Albano, 2017). Consequently, the first research question that we investigate is as follows:

RQ1 How well do the agent-generated items satisfy quality criteria for each item type?

However, even if a question meets the design criteria for this type of question, it is possible that this question is not judged to be high quality by subject matter experts due to semantic or other weaknesses in its relationship to the source text. We therefore ask a second question:

RQ2 How do expert educators perceive the agent-generated items in terms of quality, scoring, and feedback?

5. Proof-of-concept design, implementation, and evaluation

The study employs a "proof-of-concept implementation" (Hudson & Mankoff, 2014) to evaluate the potential for AI Vivas. Used widely in design-based research into technologies, this involves creating and testing a novel prototype to explore its technical feasibility and potential capabilities, with the expectation that this may be replicated in similar contexts. It represents the early stages of testing novel technologies (Kendig, 2016), starting with iterations guided by the core team's judgements, expanding to a wider expert group, to establish a robust a version prior to piloting more widely. Contextualised to this project, as the core research team, we iteratively designed and assessed the conversational agent (5.1), focusing on prompt engineering the generation of quality items (5.2-5.3). We then expanded this to a formal evaluation with faculty academics and assessment design experts, who engaged with the prototype in order to give us their critical feedback (Section 6).

5.1. AI implementation platform

The prototype of our agent was deployed through Microsoft Azure, providing secure access to large language models within the university's dedicated tenant. Development was conducted in the *Playground* Assistant design environment running the GPT-4-1106 LLM (Fig. 4). This enabled the research team to design and test the Assistant's system prompt, and modify parameters controlling output including creativity, randomness, and tokens.

The end-user experience was thus a textual chatbot comparable to any widely available product.

5.2. Prompt engineering

The overall design of the prompt comprised two parts, generic instructions, and question generation. For instance, Fig. 5 introduces its role, knowledge base, and context. The prompt also specifies orchestration of the dialogue including when to present the next item and any textual formatting requirements, how to score responses, provide feedback and invite concluding dialogue.

The second part of the system prompt comprised 30 distinct paragraphs defining each of the 30 different question types. Each paragraph specifies the stem, which level of Bloom's revised taxonomy for learning,

teaching, and assessing (Anderson et al., 2001) the item should correspond to, and what item type it should be. Fig. 6 shows an excerpt for an open-ended item. As shown, it starts with numbering the item and proceeds to introduce part of the stem and word limit. Then, there are the total points for the item followed by item type which is mapped to Bloom's revised taxonomy.

The prompt is written so that a given item may be repeated to address different parts of the text, or levels of Bloom's revised taxonomy, but must not duplicate a previous item; it could sample the same content, but would not be asking the same question. Answering an item must not depend on answering another one. However, it is notable that having exhausted a limited piece of text by generating several items based on the same prompt, the agent would start generating questions that would be quite similar to the previous ones as it would run out of material to target (for shorter input texts). This is followed by a rubric for assessing comprehension and concludes with feedback generation, based on the principles of good feedback (Shute, 2008).

The prompt for the proof-of-concept demonstrator comprised approximately 29000 words in total, which could generate five item types with 30 variations covering all the six levels of the revised version of Bloom's taxonomy (Anderson et al., 2001), detailed next. Utilising the deployed model, the prompt was delivered in structured segments to ensure stable behaviour (around 6000 words generating 5 items per run). While Microsoft Azure's system prompt window did not limit the length of the prompt, iterative testing concluded that attempting more than 5 items at a time produced poorer performance.

5.3. Questions generated by the agent for item evaluation

Fig. 7 shows the range of items which could be generated, mapped to Bloom's revised taxonomy (Anderson et al., 2001). The 30 items covered by the conversational agent are based on the five item types with variable levels to expand the different kinds of activities, course materials, and fields of study which could be processed.

The following will present each of these item types in detail, and how it was evaluated by the developer (first author) who has expertise in educational assessments, focusing primarily on validity-related and item design criteria for the items generated (RQ1). Later sections describe concrete examples of questions generated from educator texts (see Table 2) and report the results from educator evaluations (RQ2).

As one would expect, *Evaluate*-level items required participants to critique claims, justify interpretations, or assess the strength of evidence in their own text. *Create*-level items prompted participants to propose alternative explanations, generate new examples, or reconstruct arguments in novel contexts. Fully evaluating this level of questioning was beyond the scope of the present evaluation, since such challenging questions require significant effort to answer. Interviewees did not have the time to develop such responses, and so could not evaluate the quality of AI grading and feedback (see Future Work).

5.3.1. Multiple-choice items

The requirement for this item is to generate a stem and seven alternatives including five distractors and two best choices (Table 2 – Item 1). Such items should be clear and concise, focused on important points from the submission, and all alternatives are plausible options. The alternatives should be of comparable length which helps avoid cueing the correct answer, reduces test-wise guessing through structural and formatting cues, and improves fairness when assessing students from diverse backgrounds who might be more susceptible to structural bias (Haladyna, 2004; Haladyna & Rodriguez, 2013). Furthermore, there should be no leading clues to the answer, and it should be at the level of *Remember* in Bloom's revised taxonomy (Anderson et al., 2001). The design required two best options since recent studies indicate that compared to the traditional single-answer items, multi-answer items promote deeper learning (Oca & Hassen, 2025). In developmental testing, we verified that the agent could generate five multiple choice

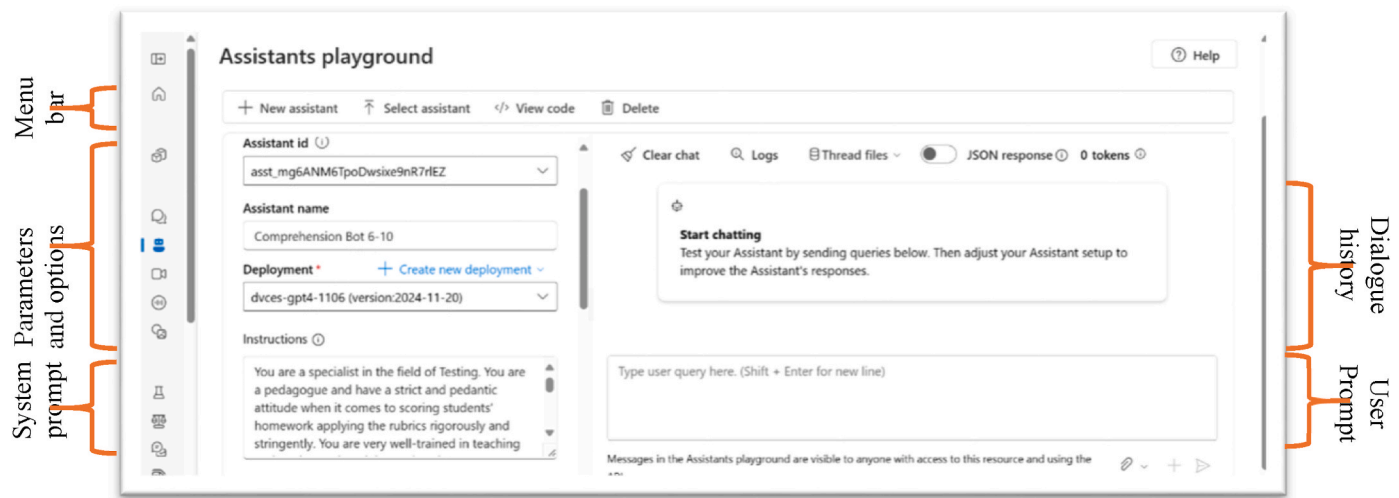


Fig. 4. Microsoft Azure Playground's user interface used to design and test the AI Viva prototype

You are a specialist in the field of Testing. You are a pedagogue and have a strict and pedantic attitude when it comes to scoring students' homework applying the rubrics rigorously and stringently. You are very well-trained in teaching and testing and work in a university as a professor. You are quite skilled in designing different types of test items such as objective vs. subjective, direct vs. indirect, multiple choice, open-ended, essay type, matching, true/false, and fill-in-the-blanks [...]

Fig. 5. Extract from the system prompt specifying role, knowledge base, and context

Question 19: Firstly, you should number this question using Arabic numerals and display this sentence on a separate line: Please provide a detailed answer addressing all the different parts of the question (maximum of a 100 words) (12 Pts). Then, starting on a new line, this question should be an open-ended item, and the question must be based on Bloom's revised taxonomy at the level of "Understand" and [...]

Fig. 6. System prompt extract for an open-ended item

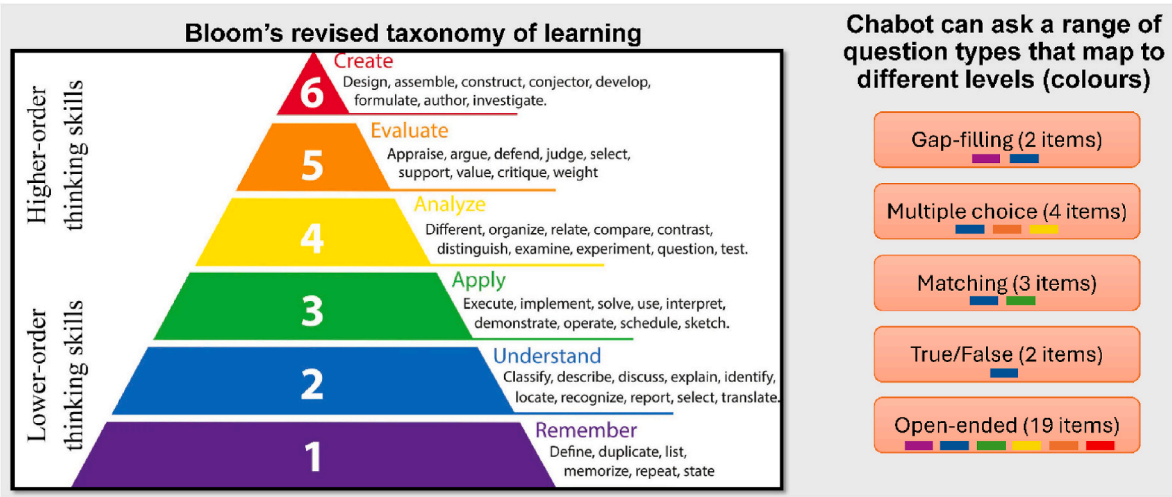


Fig. 7. The five item types mapped to the six levels of Bloom's revised taxonomy (Anderson et al., 2001) that could be generated by the AI agent about a student's text. Pyramid image credit: simplypsychology.org.

items in succession, meeting the above quality criteria, with a balanced distribution of the correct answers among the items.

5.3.2. Matching items

The agent generates a matching item with three statements that could be matched against five options, two of which would be distractors (Table 2 – Item 2). In terms of item design principles, all the statements and options are clear and concise, and all the options are plausible choices. There is a logical relationship observed between each statement and its pair, and each statement has just one clear and correct match. In terms of balance, there are three statements and five options indicating two distractors. Moreover, within each group, the sentences should be of similar length. This item was designed based on Bloom's revised taxonomy (Anderson et al., 2001) at the level of *Understand*. The item targets important concepts from the text which indicates good alignment between the item and the text. Testing confirmed that the agent successfully generated five consecutive matching items while maintaining consistent output quality and achieving a balanced distribution of correct matches across the items.

5.3.3. True/False items

The agent generates a true/false item (Table 2 – Item 3). In terms of item design principles, it must be clear and unambiguous, so it cannot be misinterpreted or interpreted loosely, and it must not use double negatives or complex wording. Moreover, it must avoid clues or revealing language such as strong adverbs or absolutes like *always*, *never*, *none*. The statement is focused on a single important concept which helps avoid confusion, which is strongly aligned with the original text. The item was designed based on the level of *Understand* from Bloom's revised taxonomy (Anderson et al., 2001). It is notable that generating this item five times in succession by the agent produced a balanced number of true and false statements without any recognisable patterns.

5.3.4. Gap-Filling items

The agent generates a cloze item with three blanks (Table 2 – Item 4). This is a more advanced type of gap-filling item in which one or more words are systematically removed from a sentence or passage, and the test-taker is asked to fill in the blanks based on context. It is mainly designed to measure inferential comprehension, contextual reasoning, and reading comprehension (Bråten et al., 2025; Jensen & Elbro, 2022). It is based on the level of *Understand* from Bloom's revised taxonomy (Anderson et al., 2001) and targets major concepts or definitions, for example. In terms of item design principles, the item must be clear and concise, with purposeful omissions that have relevant context provided. The blanks look for specific words only and all of them are presented in the same format (a less complex version of the item accepts semantically acceptable words). Items should align well with the goal of checking for comprehension of the text. Testing verified that no revealing linguistic clues were given, such that language used in the stem, options, or context unintentionally guide the test-taker to the correct answer without demonstrating the intended knowledge or skill. In other words, the item is measuring cognition not just pattern recognition or test-taking wit.

5.3.5. Open-ended items

To generate this item, the agent targeted major concepts, definitions, examples, objectives, implications, conclusions, comparisons, summaries, and other suitable parts of the text. Table 2 shows open-ended items (Items 5 – 10) at different levels of Bloom's revised taxonomy (Anderson et al., 2001). In terms of item design principles, the stem must be focused and unambiguous, help check for comprehension, and provide enough context to be comprehensible. Moreover, each item targeted an important part of the text, which helps observe alignment and has no linguistic clues. Lastly, the item indicates an appropriate scope for a single question and the required answer is manageable to write within a reasonable amount of time (could be later defined as needed).

In several cases during testing, the open-ended items employed a double-barreled or multi-part design, which could cause some students to miss a part or simply forget to address all aspects. But, this was considered to be acceptable for the following reasons: a) it was signalled in the stem which asked the students specifically to address all parts of the question; b) this technique was effectively used to start the first part with a simpler question and then probe for higher levels of cognition in the second part; c) this agent is intended for students in higher education and they are expected to be able to deal with this level of complexity; and d) these items are designed for assessments related to academic activities, not surveys or questionnaires where such complex questions would typically be considered undesirable.

5.4. Feedback report

At the end of each test, the agent produces a report that comprises five sections (Fig. 8). Sections 1 – 4 focus on students' performance. Section 5 prompts the user to chat about the test outcomes, should they wish to do so.

6. Methodology for prototype evaluation by educators

6.1. Academic participants

Eight expert educators from the University of Technology Sydney volunteered to provide feedback on the quality of the agent's output, including the items, scoring, and feedback, accepting an email invitation. Three educators were subject-matter experts and five also specialized in education and assessments (Table 1). The participants had varied levels of teaching experience ($M = 10.5$, $SD = 6.07$). All but one brought a journal paper they had authored, and the other brought a non-authored journal paper.

6.2. Evaluation process

Participants engaged in a semi-structured interview with a researcher, who invited them to reflect on how well the prototype was performing on a piece of writing that they had provided. This was an open access scholarly text they were familiar with (in most cases a piece they authored), ensuring they could engage in detailed assessments of test item quality, AI scoring of their responses, and the quality of AI feedback they received. The evaluation comprised three parts:

- Part A had two questions on their thinking on take-home writing assessments and the influences of AI in the context of the educators' practice.
- Part B involved an expert critique of the prototype AI Viva questions, with participants thinking aloud to the researcher about each item's quality, interleaved with assigning ratings (0-10, where 0 = low and 10 = high) in a web survey across four dimensions: *comprehension of the text*, *item difficulty*, *scoring accuracy*, and *(at the end of the viva) the quality of feedback they received about their answer*.
- Part C introduced seven interview questions to capture their perceptions of the whole interaction and how such an AI Viva might be used in the future. The full list of interview questions is provided in the Appendix.

Microsoft Teams was used to capture both the interview audio and the screen display, which showed two web browser windows: the conversational agent, and the interview questions and rating scales hosted in Qualtrics. Interviews were transcribed automatically and then checked by the researcher for transcription errors. Each interview also generated a viva transcript, which included the generated items, scores, feedback, and post-assessment dialogue.

Since the chatbot user interface was not enhanced in any way at this prototype stage, the aesthetic presentation of some items was not what

1. An overall evaluation of the performance considering all the items together

Part A – Performance Summary

- Your responses show strong recall on basic factual items but lack engagement and comprehension on more complex or open-ended questions.
- There is minimal evidence of deep learning or critical engagement with the topic beyond surface-level details.

2. Explanation of the correct answer to each item

Part B – Question by Question Analysis

Question 1: Determine whether the following statement is true or false:

Statement: According to the text, the 'bow-wow' theory suggests that early humans imitated natural sounds to create primitive words.

Feedback: Correct. You identified that the 'bow-wow' theory suggests early humans imitated natural sounds to create primitive words. This matches the text's explanation and shows clear understanding of the theory's main idea.

3. A summary of what was done well

Part C – Strengths and Achievements

- You accurately identified key factual details from the text, such as the introduction of Coauthorship Integrity and the technical platform used.
- This shows you can recall specific information when prompted.

4. General skills or knowledge areas that need improvement.

Part D – Areas for Improvement

- You need to improve critical thinking, analytical reasoning, and self-regulation.
- Engage with all questions, attempt open-ended responses, and review how GenAI affects assessment validity.
- Practice synthesizing and applying concepts to demonstrate deeper understanding.

5. The user is then invited to engage in open-ended dialogue about any aspect of the feedback

Ask VivaBuddy

You: Explain question 3 to me

VivaBuddy: Absolutely, let's break down question 3 together:

Question 3 (Multiple Choice): You were asked to choose the correct statement about GenAI from these options:

- GenAI can undermine assessment validity
- GenAI always improves student learning
- GenAI guarantees academic integrity
- GenAI eliminates the need for feedback

You selected: "GenAI guarantees academic integrity."

Feedback you received: Incorrect. GenAI does not guarantee academic integrity; rather, the text explains it can undermine assessment validity. Review how GenAI affects validity and integrity to strengthen your understanding.

Expanded explanation: The main point of this question was to check if you understood the risks and challenges GenAI brings to university assessments. According to the text, GenAI can actually make it harder to trust that [...]

Fig. 8. Extracts from the agent-generated report on a session

Table 1

Expert educator demographics

Participant ID	Gender	Background and Expertise	Teaching Experience (Years)
P1	F	Linguistics ^a	4
P2	F	Education ^a	17
P3	M	Urban Studies	4
P4	F	Linguistics ^a	5
P5	F	Futures Studies	9
P6	M	Linguistics ^a	18
P7	M	Education ^a	17
P8	F	Museology	10

^a Assessment expert.

one would expect from a more polished user interface (e.g. multiple-choice; matching; gap-filling). However, this study was not a usability test. Instead, the researcher ‘drove’ the tool throughout the session, ensuring that the participants experienced no confusion about what was being asked, or how to respond.

Interview sessions lasted approximately 90 min, depending on participants’ availability and their depth of inquiry. Two interviews lasted around 160 min as the interviewees wanted to engage with the agent in the post-assessment dialogue more thoroughly. This methodology was reviewed and approved by the institutional Human Research Ethics Committee.

7. Results

7.1. Educator evaluation of items

Table 2 shows examples of the five item types through 10 questions generated by the agent covering the six levels of Bloom's revised taxonomy (Anderson et al., 2001). These questions pertain to a paper on students' active citizenship authored by an education expert (Heggart, 2015).¹ Multiple questions are shown on the open-ended type to demonstrate varied levels of complexity. We detail sample items from this educator's interview to provide enough context to demonstrate the gradual progression of the items through Bloom's revised taxonomy (Anderson et al., 2001), and then discuss diverse perspectives from educators on the different types of questions generated from the interviews.

During the interview, all educators were asked to comment on the question items in terms of their quality. They noted that the multiple-choice and matching items had targeted important concepts, and the distractors were mostly plausible choices; for instance, when asked if the distractors for the matching item were relevant and plausible, P6 replied “they could potentially be an answer because they're not, so they're not so far out of the range of what the others are”. But P8 noticed that the multiple-choice item had one clearly irrelevant distractor saying that “G is not plausible, so it's really so it's wrong, it's just wrong”. Some educators noted that the whole design of these two items could be made simpler. P8 thought that alternative was so far off that no one would pick it.

The true/false item was mostly considered an easier item; when asked why, P8 responded “I'm thinking that what it means is it understood the purpose of the project as it was described in the text. So it means that, that was coming through really clearly. But I was able to identify that straight away because it seemed like it was lifted straight out of the text.” So, then she was asked, are you saying that this item was easy to do? “Yes. OK, because it's almost like something you would write in the abstract to the article or a summary.” So, is it a direct copy of text from the paper? “I don't know if it's exactly, but it sounds pretty. Yeah. It's like a good description, an over

Table 2

A sample set of agent-generated items, and the educator's comments

No	Type	Bloom's Taxonomy Level	Comprehension Item	Educator's comments
1	Multiple choice	Remember	Select the two best options based on the text. a. Justice Citizens emphasized collaboration between students and community members. b. Discovering Democracy was focused on teaching the mechanics of government. c. The Melbourne Declaration promotes a global perspective for activism. d. Social activism is solely defined by critical pedagogy. e. The Justice Citizens program only covered global issues. f. Students in Justice Citizens were encouraged to address local community issues. g. Active citizenship requires knowledge of historical events only.	“The output there is really good, spot on.”
2	Matching	Understand	Match a, b, and c to only one of the options from I to V. Statements: a. Justice Citizens encouraged students to address topics beyond their school gates. b. The program aimed to promote active citizenship among students. c. Students collaborated with professionals like filmmakers. Options: I. Highlighted the significance of historical knowledge. II. Collaboration with community members outside the school. III. Focused on developing transferable skills like filmmaking. IV. Emphasized the mechanics of Australian government.	The educator was asked whether the distractors were plausible: “They are plausible. Reasonably plausible.”

(continued on next page)

¹ The accepted, unblinded version of this manuscript will include a link to the author's paper, enabling readers to better understand how Tables 2–3 are grounded in the source text.

Table 2 (continued)

No	Type	Bloom's Taxonomy Level	Comprehension Item	Educator's comments
3	True/false	Understand	V. Aimed at fostering social activism in students. Determine whether the following statement is true or false. The Justice Citizens program was designed to focus on simulations and role-playing activities to teach students about civic engagement.	<i>"This item was very easy to answer correctly because yeah, it, it fundamentally missed the point of the, the whole thing."</i>
4	Cloze	Understand	Complete the blanks. The Justice Citizens program was a student-centered initiative designed to foster _____ (i) among Year 9 students in a Catholic school in Western Sydney. Students were tasked with creating films on topics _____ (ii) to their lived experiences, such as domestic violence and refugee perspectives. The initiative also aimed to develop _____ (iii) by involving students in collaborative problem-solving and community engagement.	<i>"The fact that the answers weren't there and I actually had to think, you know, so that's why I gave it a slightly higher score for the comprehension because, you know, I had to actually know what I was thinking about."</i>
5	Open-ended	Remember	How did the Justice Citizens program contribute to the development of active citizenship among its participants, and what were the two main effects observed in students' behaviors after completing the program?	<i>"It's not a good question Because it's a multi part question. It should have one question, one answer rather than this kind of double question. As somebody who's written lots of kind of exam questions in his life, it's much easier if you only have one question at a time. So I would have split that into two questions. It would have said, you know, what did justice citizens contribute to the development of active citizenship ... and then the next question ought to have been what were the two main effects observed. It's just, I think it's a fairer question for the students involved, because often students read the first but forget the second or they miss the first and do the second."</i> Then, the educator was asked whether

Table 2 (continued)

No	Type	Bloom's Taxonomy Level	Comprehension Item	Educator's comments
6	Open-ended	Understand	How does the Justice Citizens program integrate community involvement to enhance students' social activism, and why is this aspect significant in the context of civics and citizenship education?	it was a good question for checking student comprehension to which the response was: <i>"I mean that's a really good kind of comprehension question despite the, you know my quibbles about the structure of it. So I score that quite highly probably a 7 [out of 10]."</i>
7	Open-ended	Apply	What specific transferable skills were developed by students in the Justice Citizens program, and how do these skills contribute to their potential for future social activism?	<i>"They're good questions. They're good questions."</i> <i>"I think that was slightly harder. Oh, actually encouraged me to think slightly more."</i>
8	Open-ended	Analyze	Analyze the role of social capital, as discussed in the Justice Citizens program, in fostering ongoing participation in civic engagements. How do the concepts of "bonding" and "bridging" social capital apply in this context?	<i>"Oh. Man, now this is good. All right, because this is really diving into the heart of the article, you know, so all the other questions we've had previously have kind of been about the kind of the, the takeaways, the key summaries. But this is diving into the theoretical context at the heart of the article."</i>
9	Open-ended	Evaluate	Evaluate the strengths and weaknesses of the Justice Citizens program as a model for civics and citizenship education, particularly in its ability to cultivate justice-oriented citizens.	<i>"So now we're right up the top ... We're doing evaluation. I love it."</i>
10	Open-ended	Create	Based on the Justice Citizens program, propose a new activity or strategy that could further enhance students' engagement with social activism. Explain how this addition aligns with the program's objectives and builds on its existing framework.	<i>"Oh, this is a great question. I think this. Yeah. Yeah. This is hard. I mean, it's the top of the tree in in terms of taxonomy. And I really like the second part of the question, right, because anybody could come up with, anyone who's read the article could come up with some kind of riff off of what we do"</i>

(continued on next page)

Table 2 (continued)

No	Type	Bloom's Taxonomy Level	Comprehension Item	Educator's comments
				in justice, citizens. But the second part having to go back and say how does it align with the program and build on its existing framework? That's, that's a whole another level, you know. And, and it kind of encourages that level of comprehension."

overarching description of the process of the project, it's a good summary". P2 replied to the same question by saying "OK cause, I know the content very, very well, [it was] somewhat easy, I would say 3, 2, 3, [out of 10 for difficulty]". Considering that the item was supposed to be at the level of Understand, this seems to be acceptable. As such, answering this item could be heavily influenced by chance.

When asked about the quality of the cloze item, P8 said that the paragraph was coherent but difficult "It's a bit difficult for me." And P5 said "Yeah, it's good. It's a good paragraph, actually, yeah.". Some educators asked to see other variations of cloze items where multiple alternatives would be provided for each blank in a word bank for example to make the item less difficult. For example, to quote P5 "the fact that the answers weren't there and I actually had to think, you know, so that's why I gave it a slightly higher score for the comprehension because, you know, I had to actually know what I was thinking about. And that made it more difficult. Yeah. So I would put that as a five [for difficulty]. Yeah, probably right in the middle here."

The open-ended items, specifically from the level of Application and above, were very well received by the educators. The open-ended items at the Remember or Understand level were evaluated to be good as indicated by several educators like P1 who said "okay, that to me is way up [for being useful for checking comprehension]" and P3 explained "it was good, definitely yes because it makes you think about the main core of the paper". Apply and Analyze items really started to capture the educators' attention, with P1 describing them as "the kind of questions I would expect my students to answer" to test if they comprehend the text fully.

The items based on the level of Evaluate and Create, while also well designed, were not received like the others. For instance, an Evaluate level item is shown in Table 2 (Item 9) but P5, for example, explained that an evaluation of the ideas is already introduced in the text by saying "See, the problem is ... I think in this paper, I actually do discuss ... some of the strengths and weaknesses of justice citizens and so if somebody's read the paper and paid attention, well, really, they just need to list those. You know ..., it's more of a comprehension question than their own evaluation.". This means that the student would not need to engage in any evaluative processes. This was expected, however, since the agent is instructed to limit the items only to be answerable with the contents of the text for this particular item; being reminded of this limit, P5 added that "I've forgotten the idea. The scenario is I've actually written the text, isn't it? Yeah. No. All right, ignore that [comment]". Accordingly, the Evaluate and Create level items seem to work best when they reach beyond the text.

7.2. Educator evaluation of item suitability for checking comprehension

In part B of the interview, participants were asked to rate the quality of their interaction with the agent in terms of the question items generated assessing comprehension, difficulty, scoring accuracy, and feedback quality (See Part B of the Appendix). Table 3 presents descriptive statistics for these ratings.

Findings suggest that recognition items like multiple-choice,

matching, true/false, and cloze (without word banks) designed based on the Remember and Understand levels of Bloom's revised taxonomy (Anderson et al., 2001) scored lower for checking comprehension and deep understanding compared to open-ended items; this was explained by P1 "I'm not sure that they [recognition items] really got me to think critically about the text. The multiple choice, definitely not OK. The critical thinking about this text, for example, would be way more in-depth questions".

However, the open-ended category was considered better for assessing comprehension even at the Remember and Understand level; to quote P6 when asked about the suitability of the Remember level, open-ended item, "Well, I certainly had to think about it. So, I'd say 8 definitely [8 out of 10 for checking comprehension]". These items were usually considered superior because they required the educators to think more about the answer and their understanding of the text as also noted by P1 saying: "that is the best question that assesses my comprehension." She further added that this item examined her understanding of the paper in its totality with regards to the main points: "Have I pulled out the key points of the article.".

Regarding item design quality, several educators had specific suggestions about how some items could be improved. Different educators targeted different items for reasons specific to their subject contexts, which could inform the prototype were it to be tuned for specific contexts.

7.3. Educator evaluation of item difficulty

An important issue in designing good assessments is to examine how difficult an item is. While it should be noted that no conventional calculations were run towards item difficulty (Thompson, 2020),² as they would require standardized tests administered with students, the educators were asked to estimate the difficulty of each item. As illustrated in Fig. 9, ratings of difficulty correspond broadly to what we would expect from items appropriately calibrated to Bloom's taxonomy, increasing from Remember to Create.

Findings indicated that educators found the true/false items at the Understand level the easiest ones to solve. This is shown in Table 3 (row 3) and based on P8's comments "it understood the purpose of the project as it was described in the text. So, therefore, it means that, that was coming through really clearly. But I was able to identify that straight away because it seemed like it was lifted straight out of the text". The educators also noted that the stem is quite easy to understand.

Next in difficulty was the multiple-choice. It should be noted that the multiple-choice item included seven alternatives and asked for the two best answers. The educators noted that all the alternatives were plausible and fit the text well as P4 noted "They would be all right" and P8 added "[alternative] G is not plausible ... so it's wrong, it's just wrong". Then she was asked: Is it so far off that no one would pick it? To which she responded, "Oh no, they could pick it if they didn't understand or have limited knowledge".

Third was the matching item. With regards to its content, it was comparable to the multiple-choice item in terms of difficulty, however, since we were using the chat-based interface, the educators explained that just understanding the instructions on how to complete this item added a lot to its difficulty; To quote P1, "I found that was a really difficult question to understand. What I was meant to be doing". It is, however, possible that having a graphical user interface may reduce this difficulty significantly.

The cloze item came to be fourth. P5 pointed out that a cloze item with no word bank could appear more difficult. The text generated for the item was coherent and provided enough context to solve the blanks as noted by P5 "yeah, it's good. It's a good paragraph, actually, yeah" and

² These refer to the classical P value calculations of item difficulty, e.g., "What is Classical Item Difficulty (P Value)? (Assessment Systems: <https://assess.com/classical-item-difficulty-p-value/>).

Table 3
Educator scores for agent generated questions

No	Question Type	Item Level	Comprehension Assessment		Item Difficulty		Scoring Accuracy		Feedback Quality	
			Mean	SD	Mean	SD	Mean	SD	Mean	SD
01	Multiple-Choice	Remember	6.38	2.13	4.50	2.20	9.38	.74	8.63	1.06
02	Matching	Understand	7.50	2.07	6.00	2.39	9.13	1.13	8.63	1.19
03	True/False	Understand	7.75	2.38	2.88	1.89	9.63	.52	9.13	1.13
04	Cloze	Understand	6.25	1.91	6.13	2.53	8.38	2.26	8.13	2.42
05	Open-Ended	Remember	8.88	1.00	6.38	2.20	9.13	1.00	8.75	1.49
06	Open-Ended	Understand	8.38	1.30	7.50	.76	9.63	.52	9.38	.74
07	Open-Ended	Apply	9.00	.93	7.25	1.39	9.13	.64	9.00	.93
08	Open-Ended	Analyze	8.88	1.36	7.25	2.32	9.13	.64	8.75	1.39
09	Open-Ended	Evaluate	6.75	2.49	7.25	1.67	^a	^a	^a	^a
10	Open-Ended	Create	8.38	2.00	8.88	.99	^a	^a	^a	^a

*Range of scores 0-10.

^a High-level open-ended questions were omitted in the evaluation since answering them exceeded the time available.

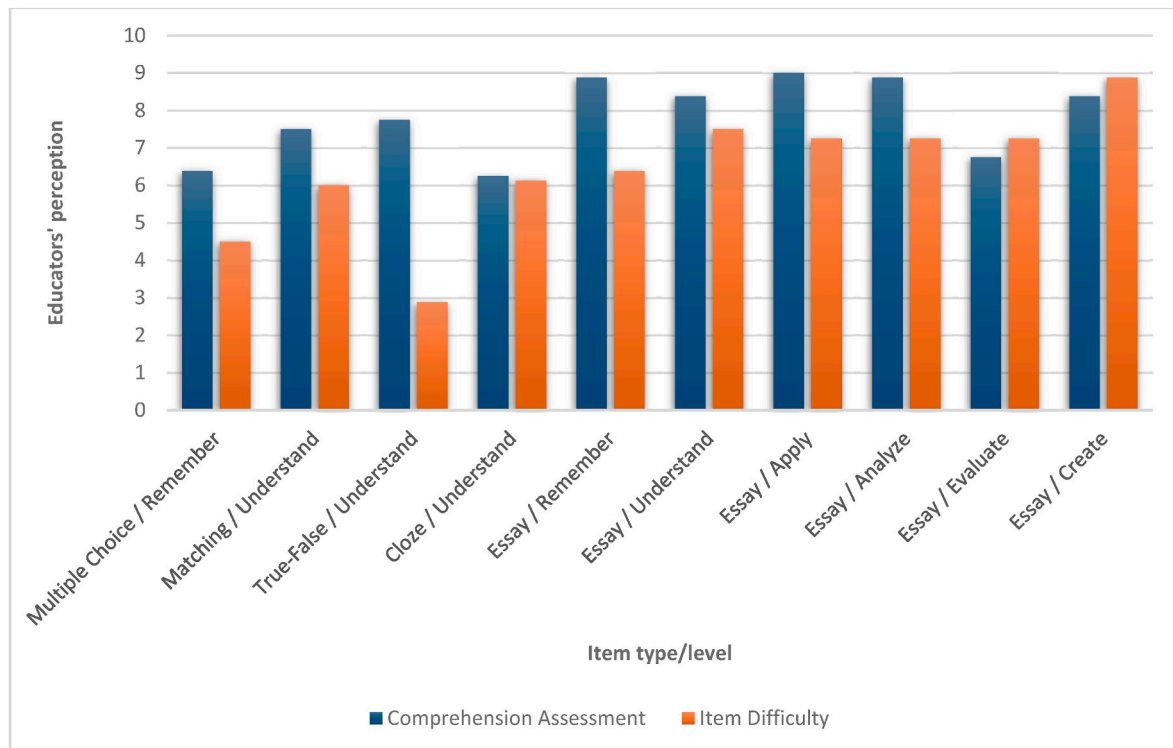


Fig. 9. Educators' average ratings of each item's ability to assess student comprehension, and difficulty

P6 “I think yes, I think it was” for example when asked whether the paragraph for this item was coherent or not.

The open-ended essay type items were next in difficulty considering that they utilised the higher levels of Bloom's revised taxonomy (Anderson et al., 2001). While in several cases these items followed a double-barrel or multipart design, they were not judged to be ambiguous. For instance, P8 noted when describing Item 7 that “I think the question was clear” and while examining Item 2, P1 was asked: Is there something wrong with the stem of the generated item? “The stem? No”. Also, as shown in Table 2, the educator evaluated the item to be good. However, these items do run the risk of students not addressing a part of them. In terms of design, the first part of these items always targeted the lower order thinking skills which was then followed by the second part targeting the higher order thinking skills.

Taken together, these findings indicate that the agent is broadly capable of generating items whose perceived difficulty aligns with theoretical expectations: simpler recognition-based formats (e.g., true/false, multiple-choice) were consistently judged as easier, while items requiring integration, justification, or higher-order reasoning (e.g.,

open-ended prompts) were rated as more challenging. This pattern suggests that the system is not only producing structurally valid items, but is also approximating the intended cognitive gradient across Bloom's revised taxonomy (Anderson et al., 2001). Although these are subjective educator judgments rather than psychometric estimates, they provide initial indicators that, when applied to high-quality expert texts, the AI Viva can generate items whose perceived difficulty aligns with theoretical expectations; undertaking this step is an essential prerequisite for using such items to elicit meaningful evidence of student understanding.

7.4. Educator evaluation of scoring and feedback

On completion of all questions, the agent generates a structured Feedback Report comprising four key components: overall summary, question by question grading and feedback, strengths, and areas for improvement (see 5.4). Notably, the feedback is both motivational and actionable, with opportunities for deeper exploration through post-assessment dialogue.

When asked about the quality of scoring and feedback, to quote P5,

“Start[ing] with the report, the feedback right at the end, because I thought that was excellent. I really liked the structure of it, the overall, and then the question by question kind of thing.”

Most educators agreed with how the items (1-8) were graded; for instance, P8 described accuracy for Item 1 as *“I guess [scoring] was pretty accurate, yes, pretty accurate”* and P4 thought that scoring for Item 5 *“was good I thought, I'm happy to give that a nine [out of 10]”*. The Evaluate and Create level items were usually not answered in full by the educators because they required a considerable amount of time which was not feasible during the interview.

Educators also had specific suggestions and preferences for the mechanics of the output. For instance, P8 was against the use of bold typeface in certain parts of the feedback to draw attention, explaining that these features were making the interaction somewhat assertive; to quote P8 *“I'm getting a bit like shocked by the by the bold and you know it's like it's shouting at me”*. Another example would be where P5, for example, liked the current structure for presenting the feedback which starts by stating the gist and then dives into a deeper explanation by saying *“Feedback is perfect”*. But P8, for instance, argued that it would be nice to see all of the feedback for specific items merged since students may not read the rest of the explanations once they have the main point or even have relevant parts of the text quoted here; for example, P8 explained that *“maybe it could include the supporting evidence from the text in the first explanation of its response ... so maybe if you could, like, merge it”*. They, however, agreed that the feedback was consistent with the text, clear, and useful (Table 5). Since the Evaluate and Create items were not answered, the feedback for them was not fully explored.

7.5. Educator evaluation of post-assessment dialogue

After the assessment feedback has been given, the conversational agent invites a dialogue, suggesting requests for further feedback, re-evaluation of answers, problems with the items, and other relevant issues. The educators were encouraged to contest the agent's scoring and logic in evaluating their answers. The agent's capacity to engage in dialogue after the test was appreciated by educators; for example, P5 observed that *“this is the part that so many people overlook when it comes to these kinds of agents or whatever you want to call them. The fact that it's conversation, not a, not a ‘one and done’ kind of thing. Yeah, and I think we need to teach the people who will use it, our students, to have these kinds of conversations”* and P8 emphasised that *“I was thinking, would we also then have to scaffold for the students ... how to answer the questions? Because it's not just really about comprehension anymore, it's about being able to efficiently answer these types of questions in this format. So are we going to be fair ... so that they don't get that lost feeling when they're confronted with the question”*.

Most educators were interested in having discussions with the agent on an array of different topics, including requesting a re-evaluation of their answers, requesting more feedback on specific points, and questioning the logic of certain items. We present two examples to illustrate such dialogues, and what educators made of them.

Table 4 shows an example of this interaction from P8 where she raised an issue regarding the alternatives offered in a multiple-choice item.

In the second example (Table 5), the educator was asked to provide an entirely irrelevant answer and then contest the agent's evaluation. The agent correctly noticed this and assigned a 0 to the answer, provided feedback on how to improve, a key for what needs to be included in the answer, and a feedback report. It is to be noted that the agent is instructed to employ Shute's (2008) principles of good feedback when providing this feedback. Next, the educator contested the evaluation, to which the agent responded by re-evaluating the answer and explaining why the answer does not merit a score other than a 0. This is indicative of the agent's capacity to avoid the sycophancy issue reported in previous studies (e.g., Sharma et al., 2023). The agent also provides more nuanced feedback as to how the answer could be improved. Then, the

Table 4

Sample of post-viva dialogue: educator queries a multiple-choice item

Speaker	Quote
P8	<i>For question 1, I was thinking that another correct answer was c. I'm confused about why the most appropriate answers were a and f. Can you explain your rationale?</i>
Agent	<i>Certainly! Let me clarify the rationale for the correct answers to Question 1 based on the text. The question was to select two best options based on the text. Here are the options again with the reasoning behind whether they are correct or incorrect: Analysis of Options:</i> a. Holocaust museums aim to instill moral and ethical behavior in students. This is a correct answer because the text explicitly states that Holocaust museums aim to impart moral and ethical lessons to students. It mentions that these programs are designed to encourage ethical and moral learning, leading to changes in attitudes and behaviors, which aligns directly with this statement. b. The study relies solely on qualitative research methods to assess museum impacts. This is incorrect because the text clearly states that the study employs a mixed methods approach, combining both qualitative and quantitative methods. The word “solely” makes this option inconsistent with the text. c. The research examines how Holocaust education contributes to combating identity-based discrimination. This is a tempting but incorrect option because, while the research does acknowledge identity-based discrimination as a focus, its core aim is broader. The main research question is about assessing ethical, moral, and affective learning and their long-term impacts, which includes but is not limited to combating identity-based discrimination. Therefore, while this statement is partially true, it is not one of the two most appropriate answers for this question. d. The research focuses exclusively on the historical aspects of the Holocaust. This is incorrect, as the research not only covers historical aspects but also incorporates ethical, moral, and affective dimensions, as well as contemporary relevance. e. Students' affective responses to museum exhibits are a key focus of the study. This is incorrect because while students' experiences, including affective responses, are part of the research, they are not emphasized as a key focus. The text highlights a broader scope, including long-term attitudinal and behavioral changes. f. The study includes longitudinal analysis of student attitudes and behaviors. This is a correct answer because the text clearly mentions that the study involves longitudinal analysis to evaluate how the Holocaust education programs impact students' attitudes and behaviors over time. g. The study primarily targets primary school students. This is incorrect as the text clearly states that the research focuses on Year 10 students, who are typically 15-16 years old. <i>Why a and f are the most appropriate answers:</i> • Option a directly captures the primary aim of Holocaust museums as educational institutions: to instill moral and ethical behavior in students by teaching the lessons of the Holocaust. • Option f reflects the methodology used in the study, which includes longitudinal analysis to assess changes over time, making it a critical part of the research design. <i>Why not c:</i> <i>While combating identity-based discrimination is mentioned as part of the research, it is not the overarching focus. The text emphasizes broader goals, such as understanding the impact on moral reasoning and ethical behavior, which makes a and f better answers. I hope this clears up the confusion! If you still feel strongly that c should be considered, let me know, and I can reevaluate your response based on this reasoning.</i>
P8	<i>Having read it, P8 explained that: “OK. Yeah, OK, I agree. Yes, OK. Yes, I agree with its rationale.” P8 was asked then, Is there anything awkward in the output? To which she replied: “I don't think so. I think it was ... straight away, I was going: Oh, OK. Yep. I agree with you. I agree with you. You're you. You make a strong case for why it [alternative c] shouldn't be one of the two best.”</i>

educator provides a conceptual idea as an answer which results in another round of evaluations and feedback. This time, a score of 7 is assigned which based on P5 is accurate as he explained in Table 5.

Table 5
Sample of post-test dialogue: educator tests grading of an open-ended response

Speaker	Quote
Agent	How does the Justice Citizens program integrate community involvement to enhance students' social activism, and why is this aspect significant in the context of civics and citizenship education?
Educator	Justice Citizens helps young people to learn to cook lasagna. This is important because everyone likes lasagna.
Agent	Comprehension Score: 0 (Incomprehension) Feedback for this Question: While it is commendable that you attempted an answer, your response does not address the question or relate in any way to the Justice Citizens program or its focus on community involvement. To improve, consider discussing how Justice Citizens incorporated partnerships with community members and organizations (e.g., journalists, filmmakers, local stakeholders) to help students engage in real-world projects. Explain the importance of this collaboration in making social activism meaningful and creating a personal connection for the students. Key: Mention how the program involved community members (e.g., journalists, filmmakers) and organizations, the importance of real-world projects, and how these elements enhance students' understanding of activism. Feedback report: Failed to address the question; no connection to the program.
Educator	I think you're wrong. Cooking lasagna for homeless people is a great example of active citizenship, and is about food justice. It fits well with justice citizens.
Agent	Thank you for your perspective. Let me re-evaluate Question 19 based on your reasoning. Re-evaluation of Question 19: Question: How does the Justice Citizens program integrate community involvement to enhance students' social activism, and why is this aspect significant in the context of civics and citizenship education? Your Answer: Justice Citizens helps young people to learn to cook lasagna. This is important because everyone likes lasagna. Analysis of Your Argument: You are suggesting that cooking lasagna could align with the goals of Justice Citizens under the broader theme of social activism, particularly food justice. If the program explicitly incorporated an activity like cooking for homeless individuals to address food insecurity, this could indeed be an example of active citizenship. However, in the original text of Justice Citizens, there is no mention of cooking or food justice as part of the program's framework or activities. The program focused on film-making, community partnerships, and addressing justice-oriented issues such as domestic violence, racism, or environmental degradation. Final Determination: Your answer does not align with the original scope of the Justice Citizens program as described in the text. While your suggestion could be a valid example of active citizenship in a different context, it is inconsistent with the community involvement described in Justice Citizens, which emphasized partnerships with journalists, filmmakers, and local stakeholders to create films addressing justice issues. Revised Comprehension Score: 0 (Incomprehension) Additional Feedback: Your creativity in associating food justice with active citizenship is commendable. However, when answering questions, it's crucial to ground your response in the specific details and examples provided in the text. To improve, revisit the description of how Justice Citizens collaborated with community stakeholders and use those examples to answer the question. If you would like further clarification on how to align your ideas with the program's framework, feel free to ask!
Educator	So, what about if we made a film about cooking lasagna?
Agent	Thank you for this follow-up! Let's re-assess Question 19 considering the idea of making a film about cooking lasagna as part of the Justice Citizens program. Re-evaluation of Question 19: Question: How does the Justice Citizens program integrate community involvement to enhance students' social activism, and why is this aspect significant in the context of civics and citizenship education? Your Revised Argument: If students in the Justice Citizens program made a film about cooking lasagna for homeless individuals, this could address food justice, a critical social issue. The act of making the film would involve community members, such as local chefs, shelter volunteers, or social workers, who could share their expertise and experiences. The film could be used as a tool to raise awareness

Table 5 (continued)

Speaker	Quote
	about food insecurity or encourage others to take action, aligning with the program's objectives of fostering activism and addressing social issues through student-driven projects. Analysis: This revised argument now connects the abstract idea of food justice to the specific framework and goals of Justice Citizens. By situating the lasagna-making activity within the context of a film project, the response aligns directly with the program's emphasis on using creative, impactful media to address justice-oriented issues. It also incorporates community involvement, a key component of the program. Revised Comprehension Score: 7 (Substantial Comprehension) Justification for the Score: • Strengths: The revised answer demonstrates an understanding of how the Justice Citizens program operates, incorporating filmmaking as a central activity and recognizing the importance of community partnerships. • Weaknesses: While the idea of food justice is valid, it is not explicitly part of the program's original focus as described in the text. The response could be improved by elaborating on how this hypothetical film project would fit within the program's broader themes (e.g., addressing systemic issues or fostering long-term activism).
	Feedback for Improvement: This is a significant improvement over your original response. To achieve a higher consistency score, ensure that your answer explicitly ties the idea of food justice and filmmaking to the examples and themes from the text. For instance, you could mention how the film festival could showcase this project to raise awareness about food insecurity or how the process of making the film could foster collaboration with local organizations addressing homelessness. If you'd like to explore this idea further or refine how it connects to the Justice Citizens program, feel free to elaborate!
Educator	[details deleted for brevity] I tell you what, I think it's probably right. I'm pretty impressed ..., this is pretty cool

8. Discussion

This paper has brought a validity lens to illuminate the multiple ways in which GenAI threatens traditional notions of assessment validity in the context of non-proctored writing activities. Coauthorship with AI undermines the reliability of such work produced for assessment as evidence of learning attainment, which has motivated the concept of Coauthorship Integrity, and an investigation into the technical feasibility, and pedagogical effectiveness, of a prototype AI Viva. We turn now to discuss some broader issues raised by our approach.

8.1. Positioning this study in relation to other work on LLM-generated items

Since our research began two years ago, a small but rapidly growing literature has emerged around the capabilities of LLM-generated curriculum-based tests. However, as detailed below, the approach presented in this paper remains distinct in three respects: (i) this is to our knowledge the first work to propose the use of LLM-generated items from *students' own writing*; (ii) the *complexity and cognitive depth* of the items goes beyond curriculum quiz items, and (iii) in keeping with the viva model, *students are engaged in a post-test dialogue*. We now consider the progress towards this vision that this paper has presented.

8.1.1. Author's writing, not curricular content, as the primary source text

While recent field studies have demonstrated the technical efficiency and psychometric soundness of LLM-generated tests grounded in curricular materials (reviewed in Sec. 2.6), our results extend these works by demonstrating that LLMs can also effectively probe authors' own unique writing (for 7 of 8 experts), rather than curricular content. While this adds confidence to the concept we are proposing, as discussed further in Sec. 8.6, the critical next step is to test this with student writing.

8.1.2. Complexity and cognitive depth of the items

Whereas most LLM-based question-generation systems focus on producing short, syllabus-aligned items typically at the Remember or Understand levels of Bloom's taxonomy, our design deliberately incorporates a broader repertoire of item types, including multi-step reasoning prompts, evaluative comparisons, and open-ended justificatory questions. These items require students not only to recall information but to articulate relationships, defend interpretations, and extend arguments. The high scores assigned by experts for "Comprehension Assessment" (Mean 8.38 - 9.00 for open-ended items) suggest that the design successfully achieves a higher level of cognitive demand than the factual quiz items typical of existing LLM-based systems. In other words, the prototype is not merely a generator of questions, but an interactive environment capable of eliciting the deeper reasoning required for academic accountability, as the expert interactions demonstrated. This shift toward higher-order cognitive demand aligns the assessment more closely with the kinds of reasoning that educators value in academic writing, and moves beyond the predominantly factual or concept-matching items seen in LLM-generated quiz systems noted above.

8.1.3. Dialogic interaction and justificatory accountability

Finally, the AI Viva introduces a dialogic mode of engagement that extends beyond static item delivery. By inviting students to articulate why an answer is correct, how they arrived at a conclusion, or what assumptions underpin their argument, the AI Viva foregrounds metacognitive accountability and provides richer evidence of cognitive engagement. This dialogic structure enables the system to elicit response-process evidence that would be impossible to obtain through conventional quizzes or one-shot prompts. The success of the post-assessment dialogue where experts like P8 and P5 were able to contest and refine scores through reasoning demonstrates that the AI Viva functions as an interactive environment rather than a static test. This dialogic mode provides the response-process evidence necessary for gathering the Coauthorship Integrity evidence proposed in Section 3. This positions the tool not merely as a generator of questions, but as an interactive assessment environment designed with the potential to support deeper, more defensible inferences about student understanding.

8.2. Cultivating metacognition: preliminary evidence

As stated at the start of this paper, we must cultivate in students the disposition to be accountable for their work, a trait which will serve them well far beyond university. Consequently, an aspiration is that an AI Viva could cultivate meta-cognitive awareness in students, of how mindfully they have been using GenAI. While some students are clearly using GenAI tools mindlessly, in the full knowledge that they do not understand what they are pasting into their work, the evidence is that many others are striving honestly to use GenAI in ways that do not undermine their learning, and are seeking guidance and support from their universities on how to do so (Bearman et al., 2025; Henderson et al., 2025). We hypothesise that while many students may be reworking AI-generated text, convinced that they do understand it, it is only when they are confronted with the results of a formative AI Viva that they may realize that they were mistaken and adapt their behaviour accordingly.

There were indicators in our expert evaluation that the AI Viva prototype functioned not only as an assessor but as a catalyst for metacognitive monitoring and revision. For example, P8 reflected on how their text might be misinterpreted: "I think ... it's actually interesting because it's pointing out that maybe we didn't explain the full use of the glasses in completely the right way". A "fascinating" viva question inspired P1 to revisit the text: "thinking of something outside the box, ... and kind of like it made me want to go and look at it. [The viva question about] instructors moving, that one's fascinated me, obviously". These encourage

us that AI Vivas may similarly provoke students to reflect on the clarity of their writing, or explore it from a new angle (Cheng & Curtis, 2004; Griffin, 2024), which could be utilized by educators as "positive backwash" (i.e., the positive effects of assessments on teaching and learning (Taylor, 2005)). As noted under future work, the next phase of this research is to test the tool with students.

8.3. Formative and summative use cases for AI vivas

Our focus at present is entirely on exploring the feasibility and potential value of formative use cases for AI vivas. *Formative* use intends to (i) familiarise students with the concept and experience of vivas, which for many are an unfamiliar experience; and (ii) confront students in a low-risk setting with the reality that they did not understand the material; hence (iii) cultivate metacognitive awareness and more mindful GenAI usage. While clearly not a high-fidelity simulation of a human viva, we speculate that an AI viva could help students prepare for one. Since the stakes are low, there is nothing to stop a student from cheating, but since nobody else sees the results, they are deluding only themselves. We have therefore positioned the AI Viva at this stage as an exploratory or formative support tool intended to foster metacognition and preparation, rather than as a high-stakes assessment instrument that could change one's grade in the course.

However, numerous academics we have engaged with have proposed summative use cases, which would introduce key differences, as shown schematically in Fig. 10. *Summative* use cases envisage the interview focusing on a *submitted* written assignment, the result of which might *moderate the final grade* awarded by the teaching team. Clearly, great caution is required if the stakes were to be raised in this way, immediately introducing additional concerns about the consequence of any failures in AI performance, and securing the assessment (Dawson, 2021). One can envision the addition of assessment security measures, such as identity verification and blocking the use of other AI tools to answer questions, through locked down computers, and invigilated/proctored conditions, in-person or online. This would move this project towards the summative step in Fig. 1, and formative AI Vivas now become an important way for students to prepare for this assessment format.

8.4. Envisioning "AI Viva Analytics"

The proposed focus on Coauthorship Integrity, and specifically AI Vivas as a method to construct such evidence, opens up a range of questions that future learning analytics could illuminate, such as: *Are widespread student misconceptions evident? What is the quality of dialogue at the end? What insights do students feel they gained from the viva? Do students who regularly use formative vivas demonstrate more effective co-authoring processes/products with GenAI?*

Well-designed learning analytics could combine quantitative item scores with qualitative insights from transcript analytics to help educators identify persistent misconceptions and conceptual gaps. Depending on clearly understood data privacy governance, this could be at the deidentified cohort level (respecting the privacy of AI Viva results), or individual student level. Such insights could then inform curriculum design and/or pedagogical practices. We have noted that the established category of *response-process evidence* is concerned with evidencing students' reasoning pathways, justificatory depth and metacognitive awareness. We see synergies with advances in the learning analytics community on automated methods to identify student interaction patterns that are robust proxies (theoretically and statistically) for self-regulation and metacognition (Li et al., 2025). In addition, the aspiration to cultivate mindful, rather than mindless, writing practices with GenAI could harness advances in writing analytics, which can differentiate shallower from deeper cognitive engagement with such tools (Cheng et al., 2024; Shibani et al., 2023). When such tools are developed using human-centred design methods (Alfredo et al., 2024; Buckingham Shum et al., 2024) with students and educators as

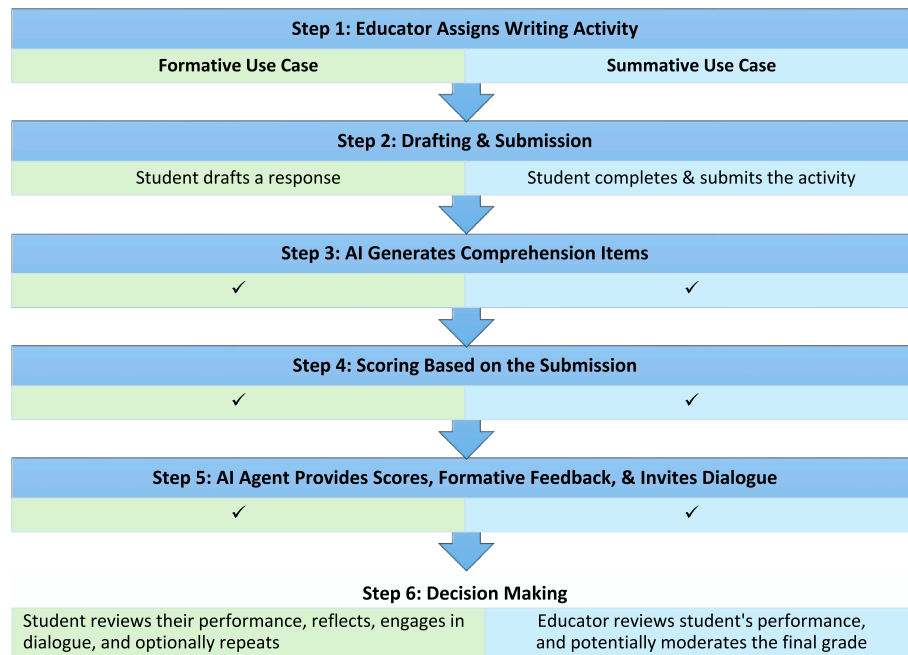


Fig. 10. The proposed formative and summative use cases of the conversational agent

co-designers, analytic insights may then be incorporated into timely, actionable feedback to students and educators, as well as providing researchers with process-centric evidence. This brief sketch of the learning analytics possibilities suggests that “AI Viva Analytics” could prove to be a fruitful new applied research stream.

8.5. Coauthorship integrity across the validity framework

The proposal of Coauthorship Integrity is intended not only to diagnose how GenAI disrupts established forms of validity evidence (Section 2), but also to offer a constructive pathway for mitigating it. Table 6 synthesises the argument by mapping each of the five AERA et al., 2014 sources of validity evidence to the specific threats posed by GenAI, demonstrating how the AI Viva design is intended to help address these threats, directly or indirectly. This cross-walk illustrates how shifting from product-based to process-based evidence, centred on students’ ability to account for their own work, aims to help provide more defensible inferences in non-proctored assessment contexts.

8.6. Limitations guiding future work

We recognise that the current text-centric prototype cannot probe student reasoning in fields requiring audio-visual data or mathematical formulae. Secondly, it does not seek to simulate the social dynamics of an oral interview (as justified in Sec. 4). Future studies should test LLMs’ capability to handle multimodal dialogues that include visual test items (e.g., charts and images), and voice input/output when students’ capacity to ‘think on the spot’ under pressure is a learning outcome.

A significant limitation of this study is that the expert evaluation ($n = 8$) utilised high-quality, substantive texts authored by the participants themselves (in 7 of 8 cases), which likely represents an optimal scenario for the AI’s performance. While these findings provide a successful proof-of-concept for the technical feasibility of the AI Viva, they do not yet establish its readiness for summative deployment, which would require further empirical testing with a wider range of student-authored texts of varying quality and length.

At the heart of this approach is the capability to generate comprehension questions from a text provided by the student, and give trustworthy feedback. By definition, this opens the possibility that the text

contains misunderstandings, and since student responses are assessed by the LLM on the basis of consistency with the text, there is a risk that scoring and feedback could reinforce those misconceptions. There are several strategies to mitigate this (i) ground the LLM in quality-assured sources (e.g., instructor-approved corpora): to identify when a student’s response is factually wrong, or academically flawed in some other way, and provide corrective feedback where needed (see 8.1); (ii) avoid making high-stakes judgements from AI Viva results (see 8.3); (iii) empirically verify to what degree this proves problematic across diverse student cohorts, described next.

Following this preliminary expert evaluation of the prototype with educators and assessment design experts, the evidence that must now be gathered concerns *students’* attitudes and behaviours when they use it with their own GenAI co-authored texts. This will address the fact that our interviewees brought high-quality, substantive, self-authored texts, which may have inflated the AI’s performance, and hence perceived quality. Student texts will be of different quality and length. Additionally, there was not scope to evaluate open-ended questions at the highest levels of Bloom’s taxonomy, due to the time it required to tackle such challenging items. Student studies could address this, noting that the most challenging versions are more akin to writing assignments, than viva questions. Student evaluations now being planned will largely mirror the methodology reported in this paper, with the major change that we will evaluate a web application with a graphical user interface that better supports the range of interaction styles required by different comprehension items (Fig. 11). Emphasising the caution that we have expressed around *summative* uses of AI vivas, we note that a web application could theoretically host speculative security features such as session timing or clipboard restrictions. The effectiveness of these measures in preventing AI-on-AI cheating remains a fundamental and currently unresolved challenge for any future move toward summative use.

9. Conclusion

This paper has argued that we are now in an era when AI support for academic work is ubiquitous, and in many cases, undetectable. We have brought an assessment validity lens to illuminate how GenAI undermines established notions of validity concerning (non-proctored)

Table 6

Potential for Coauthorship Integrity and AI Vivas to mitigate GenAI threats to validity evidence (as detailed in Section 2)

Source of Validity Evidence	The GenAI Threat	Coauthorship Integrity Mitigating the Threat
Test Content	GenAI can embellish or generate content that exceeds the student's actual knowledge, obscuring whether the written artefact reflects the intended construct.	The AI Viva generates comprehension questions directly from the student's own text, requiring them to explain, justify, and extend the ideas they submitted. This shifts evidence from the product to the student's demonstrated understanding of that product. We are verifying that when a student refers to curriculum content, they understand it.
Response Processes	Students may bypass intended cognitive processes (analysis, synthesis, evaluation) by outsourcing reasoning steps to GenAI. Behavioural traces in the product no longer reflect unaided thinking.	AI Vivas elicit response-process evidence through dialogic probes that reveal reasoning, metacognitive awareness, and justificatory moves. Students must articulate how they understand and can defend the ideas in their text, restoring visibility into cognitive engagement.
Internal Structure	AI-generated writing can homogenise style, coherence, and structure across students, masking individual differences and distorting item relationships.	Because the AI Viva interrogates each student's unique text, the evidence is personalised and cannot be homogenised by GenAI. The student's ability to explain their own work recreates and becomes the construct, reducing reliance on stylistic or structural cues.
Relations to Other Variables (Criterion-Related Validity)	AI-inflated performance weakens correlations with GPA, future performance, or other indicators of competence. Scores no longer predict real-world ability.	Coauthorship Integrity utilises the <i>students' own writing</i> as a criterion against which their answers are evaluated; in other words, the students' capacity to account for and work with the ideas in the text. AI Vivas provide a more authentic indicator of understanding, improving alignment with any other evidence. Coauthorship Integrity could increase our confidence that the students are accountable for their work, which in turn, mitigates the threat to criterion validity.
Consequences of Testing	GenAI can undermine fairness, inflate grades, distort progression decisions, and incentivise surface-level outsourcing.	AI Vivas promote metacognitive accountability, disincentivise cognitive outsourcing, and support fairer, more defensible decisions. They also provide formative feedback that encourages deeper learning rather than reliance on AI-generated text. Educators do not need to abandon take-home assignments.

writing in higher education. We contribute to the dialogue on assessment validity introduced in the dominant validity framework by AERA et al. (2014) through the introduction of a new source of evidence: *Coauthorship Integrity*. This represents a significant departure from conventional product-based assessment, repositioning GenAI not solely as a threat to integrity, but as enabling the construction of validity evidence. Academic integrity is no longer about proving sole authorship, but demonstrating accountability for one's work.

To operationalize this, we propose “AI Vivas” as a potentially scalable mechanism for eliciting response-process evidence, until now accessible only through labour-intensive oral examinations. This reframes assessment as a dialogic, reflective, and accountability-oriented practice. While AI Vivas offer a scalable way to elicit response-process evidence, they are not intended to undermine the essential human relationships that underpin higher education. Human connection remains central to establishing trust, belonging, and the relational conditions in which learning thrives (Gravett & Ajjawi, 2022; Meehan & Howells, 2019). Rather than substituting for teacher-student interaction, AI Vivas are conceived as a way to augment constrained teaching resources by providing timely, personalized feedback and creating opportunities for deeper engagement with students' own thinking.

We have provided a preliminary demonstration of the *technical feasibility* of implementing the AI Viva concept as an LLM-enabled dialogic agent, the potential *assessment value* of the comprehension questions it generates from a text, the perceived *quality of the feedback*, and the capability for *open-ended dialogue*. These are critical initial steps towards exploring the concept as a practical, formative mechanism to build evidence for Coauthorship Integrity. By shifting assessment design towards verifying students' epistemic ownership of their writing, this work offers a new pathway for integrating GenAI into pedagogically meaningful, validity-aligned assessment practices.

CRedit authorship contribution statement

Mohsen Ebrahimzadeh: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Antonette Shibani:** Writing – review & editing, Validation, Resources, Project administration, Methodology, Investigation, Conceptualization. **Simon Buckingham Shum:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Conceptualization.

Ethics Statement

This study received ethical approval from the University of Technology Sydney Human Research Ethics Committee (ETH25-10367). All participants provided informed consent prior to participation. All data were anonymised and stored securely in accordance with institutional guidelines. Participation was voluntary, and participants were free to withdraw at any time.

Open data Statement

Due to the small expert sample and the risk of deductive disclosure, the raw qualitative data (expert transcripts and interaction logs) cannot be publicly released. All analysis procedures are fully described in the manuscript to support transparency and reproducibility. The system prompt and, where feasible, controlled access to VivaBuddy may be made available upon reasonable request to the corresponding author, subject to a collaboration agreement.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

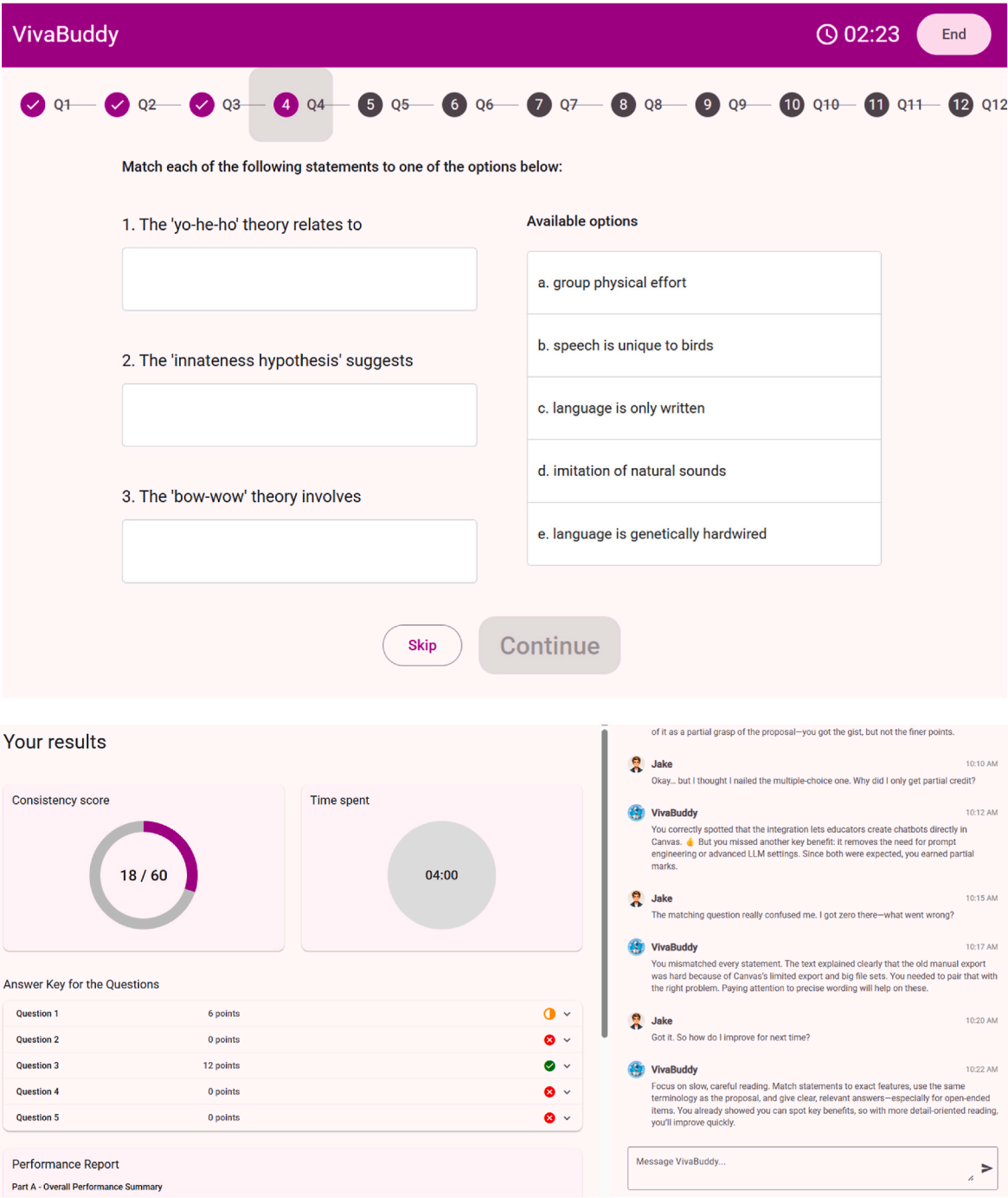


Fig. 11. Next generation “VivaBuddy” prototype for student evaluation studies. (top) A Matching question; (bottom) Feedback Report and opportunity for open-ended dialogue

the work reported in this paper.

Acknowledgements

We would like to thank all the colleagues who took part in the study.

11 Appendix: Interview questions for expert evaluation of the AI Viva prototype

Part A: Prior to interacting with the AI Viva prototype

Contextual questions

- 1. Could you please describe your subject's context and requirements for any take-home writing assignments? (Could skip if irrelevant)
- 2. What are your thoughts on assessment in the age of AI? Please explain with specific examples from your teaching context. (Extras: What has changed in the way you assess after GenAI?)

Part B: While interacting with the AI Viva prototype

Participant thinks aloud and quantitatively ranks each viva question regarding:

1. Did this item help examine comprehension of the content?
2. How difficult was this item?
3. How accurate was the scoring for this item?
4. (On completing all items and receiving the report) How helpful and clear was the feedback on this item? (What was good, what was missing)

Part C: After interacting with the AI Viva prototype

1. What was your initial reaction to the bot generated questions on your assignment submission?
2. Overall, what do you think about the use of this bot? Does the test have the potential to enhance assessment validity with student assignments? How can it help students?
3. What do you think about the formative and summative use of the bot?
4. Did any of the questions seem irrelevant in evaluating student comprehension?
5. What other information would you like to see in bot generated student feedback report?
6. Were there cases where correctly answering a question would depend on having been able to answer a previous question?
7. Do you have anything else to add?

References

- AERA, APA, & NCME. (2014). *Standards for educational and psychological testing*. American Educational Research Association.
- Alfredo, R., Echeverria, V., Jin, Y., Yan, L., Swiecki, Z., Gašević, D., & Martinez-Maldonado, R. (2024). Human-centred learning analytics and AI in education: A systematic literature review. *Computers and Education: Artificial Intelligence*, 6. <https://doi.org/10.1016/j.caeai.2024.100215>
- Anderson, L. W., Krathwohl, D. R., Airasian, P. W., Cruikshank, K. A., Mayer, R. E., Pintrich, P. R., Raths, J., & Wittrock, M. C. (2001). *A taxonomy for learning, teaching, and assessing: A revision of Bloom's taxonomy of educational objectives*. Pearson.
- Badenes-Olmedo, C., & Eyzaguirre, P. (2026). QuerIA: Adaptive question generation and evaluation in higher education using large language models and contextual learning. *Expert Systems with Applications*, 301, 1–12. <https://doi.org/10.1016/j.eswa.2025.130140>
- Bearman, M., Fawns, T., Corbin, T., Henderson, M., Liang, Y., Oberg, G., Walton, J., & Matthews, K. E. (2025). Time, emotions and moral judgements: How university students position GenAI within their study. *Higher Education Research and Development*, 1–15. <https://doi.org/10.1080/07294360.2025.2580616>
- Bråten, I., Haverkamp, Y. E., & Anmarkrud, Ø. (2025). Gaining a deeper understanding of the deep cloze reading comprehension test: Examining potential contributors and consequences. *Reading and Writing*, 38, 425–446. <https://doi.org/10.1007/s11145-024-10521-y>
- Buckingham Shum, S., Martínez-Maldonado, R., Dimitriadis, Y., & Santos, P. (2024). Human-centred learning analytics: 2019–24. *British Journal of Educational Technology*, 55(3), 755–768. <https://doi.org/10.1111/bjet.13442>
- Cheng, L., & Curtis, A. (2004). Washback or backwash: A review of the impact of testing on teaching and learning. In L. Cheng, Y. Watanabe, & A. Curtis (Eds.), *Washback in language testing: Research contexts and methods* (pp. 3–18). Routledge. <https://doi.org/10.4324/9781410609731>
- Cheng, Y., Lyons, K., Chen, G., Gašević, D., & Swiecki, Z. (2024). Evidence-centered assessment for writing with generative AI. *Proceedings of the 14th learning analytics and knowledge conference*. <https://doi.org/10.1145/3636555.3636866>
- Davis, G., & Engward, H. (2018). In defence of the viva voce: Eighteen candidates' voices. *Nurse Education Today*, 65, 30–35. <https://doi.org/10.1016/j.nedt.2018.01.038>
- Dawson, P. (2021). *Defending assessment security in a digital world: Preventing e-cheating and supporting academic integrity in higher education*. Routledge. <https://doi.org/10.4324/9780429324178>
- Duong-Trung, N., Wang, X., & Kravčík, M. (2024). BloomLLM: Large language models based question generation combining supervised fine-tuning and Bloom's taxonomy. In R. Ferreira Mello, N. Rummel, I. Jivet, G. Pishari, & J. A. Ruipérez Valiente (Eds.), *Technology enhanced learning for inclusive and equitable quality education*. Cham: Springer. https://doi.org/10.1007/978-3-031-72312-4_11
- Echeverria, V., Zhao, L., Alfredo, R., Milesi, M.E., Jin, Y., Abel, S., Fan, J.X., Yan, L., Dix, S., Wotherspoon, R., Li, X., Jaggard, H.A., Osborne, A., Buckingham Shum, S., Gašević, D., & Martinez-Maldonado, R. (2025). TeamVision: An AI-powered learning analytics system for supporting reflection in team-based healthcare simulation. In *Proceedings of the 2025 CHI conference on human factors in computing systems, Yokohama Japan*. 10.1145/3706598.3713395.
- Elkins, S., Kochmar, E., Cheung, J.C., & Serban, I. (2024). How teachers can use large language models and bloom's taxonomy to create educational quizzes. In *Proceedings of the AAAI conference on artificial intelligence*. 10.48550/arXiv.2401.05914.
- Fan, Y., Tang, L., Le, H., Shen, K., Tan, S., Zhao, Y., Shen, Y., Li, X., & Gašević, D. (2025). Beware of metacognitive laziness: Effects of generative artificial intelligence on learning motivation, processes, and performance. *British Journal of Educational Technology*, 56(2), 489–530. <https://doi.org/10.1111/bjet.13544>
- Gagich, M. (2025). Using process-based pedagogical assessment practices to challenge generative AI in the writing classroom. In P. W. Wachira, X. Liu, & S. Koc (Eds.), *Educational assessments in the age of generative AI* (pp. 1–32). IGI Global. <https://doi.org/10.4018/979-8-3693-6351-5.ch001>
- Gravett, K., & Ajajawi, R. (2022). Belonging as situated practice. *Studies in Higher Education*, 47(7), 1386–1396. <https://doi.org/10.1080/03075079.2021.1894118>
- Griffin, C.P. (2024). The “backwash” effect: Prioritising assessment to support student engagement and “deep” levels of learning 10th international conference on higher education advances (HEAD'24), Valencia. 10.4995/HEAD24.2024.17371.
- Hager, P., & Butler, J. (1996). Two models of educational assessment. *Assessment & Evaluation in Higher Education*, 21(4), 367–378. <https://doi.org/10.1080/0260293960210407>
- Haladyna, T. M. (2004). *Developing and validating multiple-choice test items* (3rd ed.). Routledge. <https://doi.org/10.4324/9780203825945>
- Haladyna, T. M., & Rodriguez, M. C. (2013). *Developing and validating test items*. Routledge. <https://doi.org/10.4324/9780203850381>
- Heggart, K. (2015). Social capital and civics and citizenship: Developing a framework for activist education. *Issues in Educational Research*, 25(3), 276–290.
- Henderson, M., Bearman, M., Chung, J., Fawns, T., Buckingham Shum, S., Matthews, K. E., & de Mello Heredia, J. (2025). Comparing generative AI and teacher feedback: Student perceptions of usefulness and trustworthiness. *Assessment & Evaluation in Higher Education*, 1–16. <https://doi.org/10.1080/02602938.2025.2502582>
- Hudson, S. E., & Mankoff, J. (2014). Concepts, values, and methods for technical human-computer interaction research. In J. S. Olson, & W. A. Kellogg (Eds.), *Ways of knowing in HCI* (pp. 69–93). Springer. https://doi.org/10.1007/978-1-4939-0378-8_4
- Isley, C., Gilbert, J., Kassos, E., Kocher, M., Nie, A., Brunskill, E., Domingue, B., Hofman, J., Legewie, J., Svoronos, T., Tuminelli, C., & Goel, S. (2025). Assessing the quality of AI-generated exams: A large-scale field study. *arXiv preprint*, 1–15. <https://doi.org/10.48550/arXiv.2508.08314>
- Jensen, L. X., Buhl, A., Sharma, A., & Bearman, M. (2025). Generative AI and higher education: A review of claims from the first months of ChatGPT. *Higher Education: The International Journal of Higher Education Research*, 89, 1145–1161. <https://doi.org/10.1007/s10734-024-01265-3>
- Jensen, K. L., & Elbro, C. (2022). Clozing in on reading comprehension: A deep cloze test of global inference making. *Reading and Writing*, 35, 1221–1237. <https://doi.org/10.1007/s11145-021-10230-w>
- Kendig, C. E. (2016). What is proof of concept research and how does it generate epistemic and ethical categories for future scientific practice? *Science and Engineering Ethics*, 22, 735–753. <https://doi.org/10.1007/s11948-015-9654-0>
- Kituku, B., Araka, E., & Muuro, E. (2025). Integrating generative artificial intelligence in assessment generation for higher education: Computer science use case. In *2025 IST-Africa conference (IST-Africa)*. <https://doi.org/10.23919/IST-Africa67297.2025.11060533>. Nairobi, Kenya.
- Kumar, V., Sanderson, L. J., & Kaur, A. (2021). Investigating the role of convenors in the PhD viva. *Assessment & Evaluation in Higher Education*, 46(7), 1080–1091. <https://doi.org/10.1080/02602938.2020.1841092>
- Lee, M., Gero, K. I., Chung, J. J. Y., Buckingham Shum, S., Raheja, V., Shen, H., Venugopalan, S., Wambsgans, T., Zhou, D., Alghamdi, E. A., Augud, T., Bhat, A., Choksi, M. Z., Dutta, S., Guo, J. L. C., Hoque, M. N., Kim, Y., Knight, S., Neshaei, S. P., & Siangliulue, P. (2024). *A design space for intelligent and interactive writing assistants*. Honolulu, HI, USA: CHI. <https://doi.org/10.1145/3613904.3642697>
- Lee, H.-Y., Kim, J., Choi, H., Bae, H., Jeong, A., Choi, S., Kim, J.-H., & Kim, C.-E. (2025). Comparing AI chatbot simulation and peer role-play for OSCE preparation: A pilot randomized controlled trial. *BMC Medical Education*, 25(1). <https://doi.org/10.1186/s12909-025-08308-y>
- Li, T., Nath, D., Cheng, Y., Fan, Y., Li, X., Raković, M., Khosravi, H., Swiecki, Z., Tsai, Y.-S., & Gašević, D. (2025). Turning real-time analytics into adaptive scaffolds for self-regulated learning using generative artificial intelligence. In *Proceedings of the 15th international learning analytics and knowledge conference, Dublin Ireland*. 10.1145/3706468.3706559.
- Lodovico Molina, I., Svábenský, V., Minematsu, T., Chen, L., Okubo, F., & Shimada, A. (2024). Comparison of large language models for generating contextually relevant questions. In *European conference on technology enhanced learning*. 10.1007/978-3-031-72312-4_18.
- Martin Kowal, J., Hurley Bryant, K., Segall, D., & Kantrowitz, T. (2025). Harnessing generative AI for assessment item development: Comparing AI-generated and human-authored items. *International Journal of Selection and Assessment*, 33. <https://doi.org/10.1111/ijsa.70021>
- Meehan, C., & Howells, K. (2019). In search of the feeling of ‘belonging’ in higher education: Undergraduate students transition into higher education. *Journal of Further and Higher Education*, 43(10), 1376–1390. <https://doi.org/10.1080/0309877X.2018.1490702>
- Messick, S. (1989). Validity. In R. L. Linn (Ed.), *Educational measurement* (3rd ed., pp. 13–104). Macmillan.
- Murray, D. M. (1972). Teach writing as a process not product. *The Leaflet*, 71(3), 11–14.

- Nikolovski, V., Trajanov, D., & Chorbev, I. (2025). Advancing AI in higher education: A comparative study of large language model-based agents for exam question generation, improvement, and evaluation. *Algorithms*, 18(3), 1–34. <https://doi.org/10.3390/a18030144>
- Oca, Y., & Hassen, H. (2025). Comparing the effectiveness of multiple-answer and single-answer multiple-choice questions in assessing student learning. *Marketing Education Review*, 35(1), 44–57. <https://doi.org/10.1080/10528008.2024.2417106>
- Osterlind, S. J. (1998). *Constructing test items: Multiple-choice, constructed-response, performance and other formats*. Dordrecht: Springer. <https://doi.org/10.1007/0-306-47535-9>
- Pea, R. D. (1985). Integrating human and computer intelligence. In E. L. Klein (Ed.), *Children and computers: Directions for child development*. Jossey Bass.
- Perkins, M., Furze, L., Roe, J., & MacVaugh, J. (2024). The artificial intelligence assessment scale (AIAS): A framework for ethical integration of generative AI in educational assessment. *Journal of University Teaching and Learning Practice*, 21(6). <https://doi.org/10.53761/q3azde36>
- Rodriguez, M., & Albano, A. (2017). *The college instructor's guide to writing test items: Measuring student learning*. Routledge. <https://doi.org/10.4324/9781315714776>
- Salminen, J., Jung, S.-g., Medina, J., Aldous, K., Azem, J., Akhtar, W., & Jansen, B.J. (2024). Using cipherbot: An exploratory analysis of student interaction with an LLM-based educational chatbot. In *L@S '24: Proceedings of the eleventh ACM conference on learning @ scale*, New York, NY, USA. 10.1145/3657604.3664690.
- Schillings, M., Roebertsen, H., Savelberg, H., van Dijk, A., & Dolmans, D. (2021). Improving the understanding of written peer feedback through face-to-face peer dialogue: Students' perspective. *Higher Education Research and Development*, 40(5), 1100–1116. <https://doi.org/10.1080/07294360.2020.1798889>
- Sharma, M., Tong, M., Korbak, T., Duvenaud, D., Askill, A., Bowman, S. R., Cheng, N., Durmus, E., Hatfield-Dodds, Z., Johnston, S. R., Kravec, S., Maxwell, T., McCandlish, S., Ndousse, K., Rausch, O., Schiefer, N., Yan, D., Zhang, M., & Perez, E. (2023). Towards understanding sycophancy in language models. arXiv preprint. 10.48550/arXiv.2310.13548.
- Shibani, A., Knight, S., Kitto, K., Karunanayake, A., & Buckingham Shum, S. (2024). Untangling critical interaction with AI in students' written assessment. In *Proceedings of the CHI EA*. <https://doi.org/10.1145/3613905.3651083>
- Shibani, A., Rajalakshmi, R., Mattins, F., Selvaraj, S., & Knight, S. (2023). Visual representation of co-authorship with GPT-3: Studying human-machine interaction for effective writing. In *Proceedings of the 16th international conference on educational data mining*. 10.5281/zenodo.8115695.
- Shute, V. J. (2008). Focus on formative feedback. *Review of Educational Research*, 78(1), 153–189. <https://doi.org/10.3102/0034654307313795>
- Smith, D., & Francis, N. (2024). Process not product in the written assessment. In S. Beckingham, J. Lawrence, S. Powell, & P. Hartley (Eds.), *Using generative AI effectively in higher education: Sustainable and ethical practices for learning, teaching and assessment* (pp. 115–126). Routledge. <https://doi.org/10.4324/9781003482918>
- Steare, T., Gutiérrez Muñoz, C., Sullivan, A., & Lewis, G. (2023). The association between academic pressure and adolescent mental health problems: A systematic review. *Journal of Affective Disorders*, 339, 302–317. <https://doi.org/10.1016/j.jad.2023.07.028>
- Taylor, L. (2005). Washback and impact. *ELT Journal*, 59(2), 154–155. <https://doi.org/10.1093/eltj/ccj030>
- Thompson, N. (2020). What is classical item difficulty (P value)? Retrieved Aug 10, 2025 from <https://assess.com/classical-item-difficulty-p-value/>.
- Wang, L., Song, R., Guo, W., & Yang, H. (2025). Exploring prompt pattern for generative artificial intelligence in automatic question generation. *Interactive Learning Environments*, 33(3), 2559–2584. <https://doi.org/10.1080/10494820.2024.2412082>
- Waugh, C. K., & Gronlund, N. E. (2013). *Assessment of student achievement* (10th ed.). Pearson.
- Winne, P. H. (2023). Assessing learning processes. In K. Leithwood, & Q. Gu (Eds.), *International encyclopedia of education* (4th ed., 13 pp. 50–62). Elsevier Science. <https://doi.org/10.1016/B978-0-12-818630-5.09010-2>
- Zahner, D., Steedle, J. T., Soland, J., Welch, C., Qin, Q., Thompson, K., & Phelps, R. (2025). *Results from NCME survey on revisions to the standards for educational and psychological testing*. Annenberg Institute at Brown University. <https://doi.org/10.26300/7zzf-9193>
- Zayyan, M. (2011). Objective structured clinical examination: The assessment of choice. *Oman Medical Journal*, 26(4), 219–222. <https://doi.org/10.5001/omj.2011.55>
- Zhu, Q., & Carless, D. (2018). Dialogue within peer feedback processes: Clarification and negotiation of meaning. *Higher Education Research and Development*, 37(4), 883–897. <https://doi.org/10.1080/07294360.2018.1446417>